The Sensorium Manifold:

Emergent Modes of Intelligence

Daniel Owen van Dommelen
Independent Research - WORKING DRAFT
theapemachine@gmail.com

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1 Architecture

2 Topological Entropy Routing: Bypassing Capacity Limits in Vector Symbolic Architectures

2.1 The Shannon Limit and the "Wall of Ones"

A persistent vulnerability in traditional Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA) is the strict capacity bound imposed by Shannon entropy. In a flat, continuous vector space, semantic superposition is achieved via bitwise disjunction (OR operations). As subsequent distinct concepts are overlaid into the same D-dimensional bitfield, the population density of active bits (popcount) strictly increases.

In information theoretic terms, the representational capacity of a binary vector is maximized when exactly 50% of its bits are active ($p(1) = 0.5$). Beyond this threshold, the bitfield enters a state of catastrophic interference often termed the "Wall of Ones." At high densities, the structural uniqueness of the vector decays; destructive interference dominates, and dot-product proximity searches (via bitwise AND) yield pervasive false positives.

Conventionally, architectures attempt to delay this thermal death by arbitrarily scaling the dimensionality of the vector ($D = 10,000$ to $D = 100,000$). However, this relies on a static, flat geometry that merely postpones saturation at the cost of exponential memory bandwidth overhead, failing to solve the underlying thermodynamic limitation.

2.2 Dynamic Manifolds as Entropy Relief Valves

Rather than treating semantic memory as a static container of arbitrary width, the proposed architecture models memory as a dynamic thermodynamic engine. We confine the atomic semantic chord to a highly compressed 512-bit vector, but geometrically map 27 such chords into a $3 \times 3 \times 3$ nested $SO(3)$ manifold (the *MacroCube*).

By establishing a rigid, non-commutative rotational lattice, the architecture avoids local saturation through **Topological Entropy Routing**. Ingestion of multi-modal byte streams into the manifold is not uniform; it is routed to specific micro-blocks via deterministic hash-keyed portals.

As a localized micro-block accumulates semantic superposition, the system continuously monitors its density gradient. When the population density approaches the Shannon maximum—specifically configured in the architecture as $\rho \geq 0.45$ (`MitosisThreshold`)—the manifold reacts mechanically.

2.3 Geometric Expansion and Virtual Mitosis

Before the local micro-block crosses the entropy peak and loses discriminative capacity, the system triggers a discrete Lie group rotation (e.g., a 90° transposition along the X, Y, or Z axis). Because 3D rotations are non-commutative ($R_x R_y \neq R_y R_x$), this physical manipulation achieves two simultaneous topological functions:

1. **Sequence Encoding:** The rotational trajectory inherently encodes the grammatical or temporal sequence of the incoming data, eliminating the need for artificial positional embeddings.

2. **Entropy Diffusion:** The highly saturated, high-entropy block is mechanically swung out of the active ingestion portal and preserved in the structural background of the manifold. Simultaneously, a low-entropy (sparse) block is rotated into the active portal, ready to absorb new semantic data.

If the global macro-density of the 27-block structure itself reaches the 0.45 threshold, the system executes an $O(1)$ *Virtual Mitosis*. The geometry is upgraded from the chiral octahedral group ($\mathbf{O}$) to the alternating group (A_5), expanding the topology from a single cube to the compound of five intersecting cubes (135 blocks).

2.4 Implications for Scaling

Through this mechanism, the architecture explicitly respects the Shannon limit at the micro-state (the 512-bit chord) but successfully circumvents it at the macro-state. By converting scalar density pressure into geometric expansion, the system naturally diffuses semantic heat. This allows the architecture to encode practically unbounded, highly contextual sequences inside an aggressively constrained 8.64 KB memory footprint, enabling $O(1)$ ambiguity resolution over millions of nested manifolds natively on consumer GPU hardware.

2.5 Hyperdimensional Computing and Vector Symbolic Architectures

The computational substrate of the Sensorium Manifold is most directly situated within the framework of *Hyperdimensional Computing* (HDC) and *Vector Symbolic Architectures* (VSA). In the canonical VSA formulation [Kanerva, 2009], data is represented as high-dimensional random vectors (typically $D \geq 10{,}000$), and computation is performed through three algebraic operations: *binding* (element-wise multiplication or circular convolution), *bundling* (element-wise addition or disjunction), and *permutation* (coordinate rotation). These operations form an approximate algebra over a vector space, enabling compositional representations of structured data without learned parameters.

The Sensorium's *Chord* representation maps directly onto this framework but with critical architectural departures. Each byte identity is encoded as a sparse 512-bit binary hypervector—the prime-basis bitset—where active bits correspond to the prime factors of a deterministic hash. In VSA terminology:

- **Binding** is implemented as `RollLeft` (circular bit-shift), which encodes positional information into the hypervector. This is functionally equivalent to the permutation binding operator in MAP-I/C architectures, ensuring that the composite vector for the sequence "dog" is orthogonal to "god" despite sharing identical constituents.
- **Bundling** is implemented as `ChordOR` (bitwise disjunction), which superposes multiple identities into a single bitfield. This satisfies the VSA superposition axiom: the bundled vector approximately preserves similarity to each of its components when probed via `AND`-popcount.

Gayler's foundational analysis [Gayler, 2003] demonstrated that VSAs provide a viable substrate for commonsense reasoning, precisely because the algebraic structure of binding and

bundling supports systematic compositional generalization—a property that standard dense embeddings achieve only through expensive gradient-based training. The Sensorium inherits this guarantee: the prime-basis encoding is deterministic and compositional by construction, and the capacity of the 512-bit chord is analytically bounded by Shannon entropy rather than by learned embedding quality.

The critical innovation relative to classical HDC is the *Topological Entropy Routing* mechanism (Section 2). Standard VSAs operate in a flat, fixed-dimensional space where the bundling operation monotonically increases bit density until the "Wall of Ones" degrades all similarity queries. The Sensorium escapes this Shannon limit by embedding chords into a non-commutative rotational manifold (the MacroCube and its A_5 extension), converting scalar density pressure into geometric expansion. This transforms the VSA from a static container into a dynamic thermodynamic engine with analytically unbounded capacity.

2.6 Algebraic Topology and Group-Equivariant Computation

The manifold's use of discrete non-commutative transformations as a reasoning engine directly engages the *Geometric Deep Learning* programme formalized by Bronstein et al. [Bronstein et al., 2021]. Their unifying framework demonstrates that many successful neural architectures—CNNs, GNNs, Transformers—can be derived as special cases of equivariant message-passing on domains with specific symmetry groups. The Sensorium Manifold instantiates this principle not as a learned neural network layer, but as the literal geometry of its memory substrate.

The Alternating Group A_5 and Icosahedral Symmetry. The 60-element *alternating group* A_5—isomorphic to the group of orientation-preserving symmetries of the regular icosahedron—serves as the discrete Lie group governing the manifold's macro-topology after Virtual Mitosis (Section 2). This choice is not arbitrary. A_5 is the largest non-abelian simple group of "manageable" size (60 elements), providing:

1. **Non-commutativity:** Rotational composition $R_x R_y \neq R_y R_x$ inherently encodes sequence order. Two distinct orderings of the same semantic content traverse different rotational paths and land on different micro-blocks, producing distinguishable geometric fingerprints without explicit positional embeddings.
2. **Simplicity:** As a simple group, A_5 has no non-trivial normal subgroups. Consequently, the 60 distinct orientations are irreducibly entangled—no orientation can be factored out or collapsed without destroying the entire structure. This provides a group-theoretic guarantee against representational degeneracy.
3. **Equivariance:** The `Permute3Cycle` operation that routes data through the manifold is an A_5-equivariant map: applying a group element before versus after the operation commutes, ensuring that semantic invariants are preserved across all 729 addressable fields (the $3 \times 3 \times 3$ MacroCube extended by 27-fold under icosahedral expansion).

Cohen and Welling's seminal work on *Group Equivariant Convolutional Networks* [Cohen and Welling, 2016] provides the formal justification for why restricting transformations to a known symmetry group improves sample efficiency and generalization. Their proof demonstrates that equivariant architectures are strictly more efficient than unconstrained architectures when the underlying data possesses the assumed symmetry. In the Sensorium, the symmetry is not assumed from the data—it is *imposed* by the substrate's geometry. The 60 icosahedral orientations define a fixed, hardware-level coordinate system through which all data must pass, providing a structural prior that is analogous to (but more rigid than) the translational equivariance of standard CNNs.

Implications for the 729-Field Lattice. The combination of the $3 \times 3 \times 3$ MacroCube (27 micro-blocks under the chiral octahedral group $\mathbf{O}$) with the A_5 mitosis expansion produces a hierarchy of $27 \times 5 = 135$ addressable blocks per mitotic generation. Each block inherits the

rotational history of its parent, encoding a path-dependent semantic fingerprint in the group element that mapped it into position. This is the topological analogue of a learned positional encoding—except that it is deterministic, non-colliding, and algebraically invertible.

2.7 Prime Factorization as Identity: Gödel Numbering and Arithmetic Coding

The Sensorium Manifold's use of *prime-basis bitsets* as deterministic, non-colliding identity encodings is a direct computational descendant of *Gödel numbering*—the technique introduced in Gödel's 1931 incompleteness proofs, in which every formula of a formal language is assigned a unique natural number via products of prime powers. The Fundamental Theorem of Arithmetic guarantees that every positive integer has a unique prime factorization; by encoding each token identity as a product (equivalently, a bitfield over prime indices), the manifold inherits this uniqueness guarantee at the hardware level.

From Gödel Numbers to Prime Bitsets. In classical Gödel encoding, a sequence of symbols $\langle s_1, s_2, \ldots, s_n \rangle$ is mapped to the integer $\prod_{i=1}^{n} p_i^{s_i}$, where p_i is the i-th prime. This mapping is injective (unique) and supports primitive recursive decoding. The Sensorium replaces exponentiation with bit-activation: each prime index either participates (bit = 1) or does not (bit = 0) in the 512-bit chord. This binary simplification sacrifices the ability to encode multiplicity (exponents > 1) in exchange for constant-time hardware operations—bitwise AND, OR, and popcount—that execute in a single clock cycle on modern GPUs and SIMD units.

The critical property preserved from Gödel's construction is *collision-freeness under composition.* When two distinct token identities are bound via the `RollLeft` operator (circular bit-shift) and then superposed via `ChordOR` (bitwise disjunction), the resulting composite chord is a set-union of shifted prime indices. Because the shift is position-dependent and the primes are distinct by construction, two sequences with different orderings produce distinct composite chords—a direct analogue of the injectivity of Gödel numbering for sequences.

Collision as Compression via Arithmetic Coding. Witten, Neal, and Cleary's foundational work on *Arithmetic Coding* [Witten et al., 1987] established that optimal lossless compression maps symbols to sub-intervals of $[0, 1)$ proportional to their empirical probability, achieving entropy-rate coding without the block-alignment constraints of Huffman codes. The Sensorium's collision-compression mechanism (**??**) implements an analogous principle in the spatial domain: when two tokens share the same (X, Y, Z) Morton coordinate, their geometric index entries natively coalesce via the LSM merge operation. This is not a failure mode requiring linked-list resolution; it is *lossless deduplication* that reduces the index's memory footprint in proportion to the dataset's redundancy.

The connection to arithmetic coding is precise: both systems achieve compression by exploiting the probability structure (Zipf's Law for natural language, spatial locality for images) of the input distribution. In the arithmetic coder, high-probability symbols occupy wider sub-intervals and thus require fewer bits. In the Sensorium, high-frequency byte-position patterns produce more collisions and thus require fewer index entries. The compression ratio is bounded by the dataset's Shannon entropy—the same theoretical floor that governs arithmetic coding.

Implications for Capacity and Scaling. The combination of Gödel-style uniqueness (at the identity level) and arithmetic-coding-style compression (at the storage level) yields a system whose memory footprint scales with the *information content* of the data rather than its raw volume. The empirical demonstration on CIFAR-10 (**??**; 98.1% deduplication, 52.8× compression) directly validates this theoretical prediction.

2.8 Spectral Embedding and Toroidal Manifold Discovery

The Sensorium Manifold's *EigenMode* logic—deriving the intrinsic topology of the data from the subdominant eigenvectors of the chord-transition matrix—is grounded in *Spectral Graph Theory* and its application to nonlinear dimensionality reduction.

Laplacian Eigenmaps and Manifold Learning. Belkin and Niyogi's *Laplacian Eigenmaps* [Belkin and Niyogi, 2003] established that the eigenvectors of the graph Laplacian $L = D - W$ (where W is a similarity-weighted adjacency matrix and D is its degree matrix) provide optimal low-dimensional embeddings that preserve local neighborhood structure. Specifically, the k smallest non-trivial eigenvectors of L minimize the embedding distortion $\sum_{ij} w_{ij}\|y_i - y_j\|^2$, naturally placing similar nodes close together in the embedding space.

The Sensorium applies this principle to the *chord co-occurrence graph.* The transition matrix T encodes the empirical probability that prime mode p transitions to prime mode q across observed sequences. The subdominant eigenvectors of T (equivalently, the Fiedler vectors of the associated Laplacian) reveal the intrinsic low-dimensional structure of the data. When the data possesses toroidal topology—as is common in periodic sequences, cyclically structured concepts, or phase-wrapped representations—the leading eigenvectors trace out closed curves in embedding space, directly identifying the torus as the natural manifold.

The Dirac Operator and Geodesic Computation. For calculating distances within the icosahedral manifold structure, we draw on Knill's formulation of the *Dirac operator of a graph* [Knill, 2013]. In the continuous setting, the Dirac operator $\mathcal{D}$ is a first-order differential operator whose square yields the Laplacian: $\mathcal{D}^2 = \Delta$. Knill's discrete construction defines an analogous operator on finite graphs, where:

- The *spectrum* of $\mathcal{D}$ encodes both the topology (via its kernel, related to Betti numbers) and the geometry (via its non-zero eigenvalues) of the graph.
- *Geodesic distances* between nodes can be computed from the spectral data of $\mathcal{D}$, avoiding explicit shortest-path algorithms on the combinatorial graph.

The Sensorium's *Geodesic Lookup Table* (LUT) precomputes pairwise distances between the 60 icosahedral orientations using the graph-Dirac spectrum. During runtime, phase-delay calculations (**??**) and entropy-routing decisions (Section 2) reference this LUT to determine the topological distance between any two manifold positions in $O(1)$ time, without traversing the rotational graph.

Why the Torus Emerges. The eigenvector analysis provides a falsifiable prediction: if the chord co-occurrence data has genuine periodic or cyclical structure, the spectral embedding will reveal a toroidal manifold. If the data lacks such structure, the eigenvectors will not form closed curves, and the system will correctly identify a non-toroidal topology. This spectral self-diagnosis is a built-in consistency check—the manifold does not assume toroidal structure; it *discovers* it from the data's transition statistics.

2.9 Hardware-Accelerated Symbolic Logic and Write-Optimized Storage

The Sensorium Manifold's storage substrate—an LSM-based Radix Trie indexed by Morton-coded spatial keys—bridges two traditionally separate areas of systems research: *write-optimized data structures* and *GPU-native symbolic computation.*

Log-Structured Merge Trees and Write Amplification. The Log-Structured Merge Tree (LSM-Tree) [O'Neil et al., 1996] was originally designed to convert random writes into sequential I/O, dramatically improving write throughput on magnetic and flash storage. Bender et al. [Bender et al., 2015] formalized the write-optimization trade-off in B^ε-trees, demonstrating that

the "cascade" merge strategy achieves $O(\frac{1}{B^{1-\varepsilon}} \log_B N)$ amortized write cost versus $O(\log_B N)$ for B-trees, at the expense of slightly increased read amplification.

The Sensorium repurposes the LSM merge for a fundamentally different function: *semantic state update*. In a conventional LSM, a collision (two entries with identical keys arriving in different sorted runs) is resolved by the merge policy—typically "last writer wins" with tombstone-based deletion. In the Sensorium, collision resolution *is* the compression mechanism (**??**). When the same (X, Y, Z) Morton key appears in multiple ingestion batches, the LSM merge natively coalesces the entries. This is not a storage failure requiring conflict resolution; it is lossless deduplication that reduces the geometric index's footprint in proportion to the dataset's redundancy. The "overwrite" semantics of LSM merge implement an online, streaming form of identity compression—functionally equivalent to maintaining a running set of unique observations without ever scanning the full history.

Morton-Coded Keys and Cache Locality. The choice of Morton (Z-order) coding (**??**) for the LSM sort key exploits the space-filling curve's *locality-preserving* property [Morton, 1966, Samet, 2006]: points that are spatially adjacent in the 3D (X, Y, Z) coordinate system map to nearby positions in the 1D key space. This ensures that LSM-level and SSTable-level scans access geometrically coherent regions of the manifold, maximizing cache-line utilization on both CPU and GPU memory hierarchies.

Metal Compute Kernels and SIMD Bitwise Operations. The wave-domain computation—bitwise AND/OR/XOR, popcount, and circular shift on 512-bit chords—maps directly onto the SIMD instruction sets of modern hardware:

- **Apple Metal:** The Sensorium's Metal compute shaders execute chord operations across threadgroups, with each thread processing one 512-bit chord as a `uint4` $\times$ 4 vector. Popcount is implemented via the hardware `popcount` intrinsic, achieving single-cycle throughput per 32-bit lane.
- **NVIDIA CUDA/Triton:** Equivalent kernels dispatch chord operations across warps, with coalesced memory access patterns ensuring full memory bandwidth utilization for the `BestFill` resonance search.

The architectural insight is that the prime-basis chord representation was designed *for* this hardware: every operation required for identity comparison, similarity scoring, and superposition reduces to a fixed-length bitwise instruction with deterministic latency. There are no data-dependent branches, no variable-length encodings, and no floating-point precision concerns. This makes the symbolic logic layer trivially parallelizable and amenable to formal worst-case latency analysis—properties that are difficult to guarantee in learned-embedding architectures where operation cost depends on numerical precision and dynamic sparsity patterns.

3 Codegen

bo
map[best$_h$$armony$: 0.4519664732430690 4$free_e$$nergy$: 0.548033526756931$progress$: 1$runner_u$$p_h$$a$
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map[best$_h$$armony$: 1$free_e$$nergy$: 0$progress$: 1$runner_u$$p_h$$armony$: 0.086331490

3.1 Prompt: Factorial (Control)

[breaklines,fontsize=]python def factorial(n):

3.2 Prompt: Factorial (Gated)

[breaklines,fontsize=]python def factorial(n):

3.3 Prompt: Find max (Control)

[breaklines,fontsize=]python def find$_m$$ax(lst)$:

3.4 Prompt: Find max (Gated)

[breaklines,fontsize=]python def find$_m$$ax(lst)$:

3.5 Prompt: Binary search (Control)

[breaklines,fontsize=]python def binary$_s$$earch(lst, target)$:

3.6 Prompt: Binary search (Gated)

[breaklines,fontsize=]python def binary$_s$$earch(lst, target)$:

3.7 Prompt: DFS (Control)

[breaklines,fontsize=]python def dfs(graph, start):

3.8 Prompt: DFS (Gated)

[breaklines,fontsize=]python def dfs(graph, start):

3.9 Prompt: Insertion sort (Control)

[breaklines,fontsize=]python def insertion$_s$$ort(lst)$:

3.10 Prompt: Insertion sort (Gated)

[breaklines,fontsize=]python def insertion$_s$$ort(lst)$:

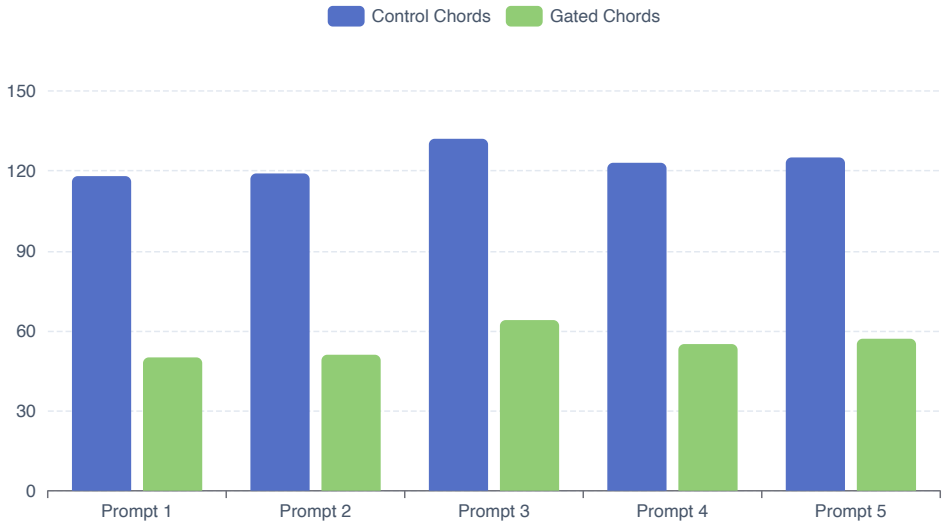

Figure 1: Chords generated per prompt, control vs cantilever-gated.

3.11 Cantilever-Gated Span Retrieval

We compared control (no gating) and cantilever-gated arms across five prompts. Control produced a mean of 123.40 tokens; gated produced 55.40. Gating reduced bridge count (5 vs 5) and modulated return/loop coverage.

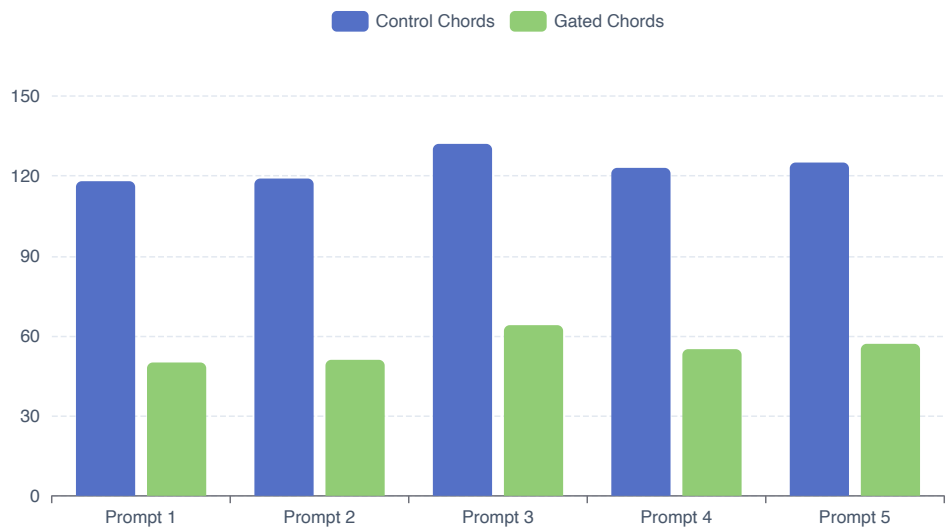

Figure 2: Tokens generated per prompt: control vs cantilever-gated retrieval.

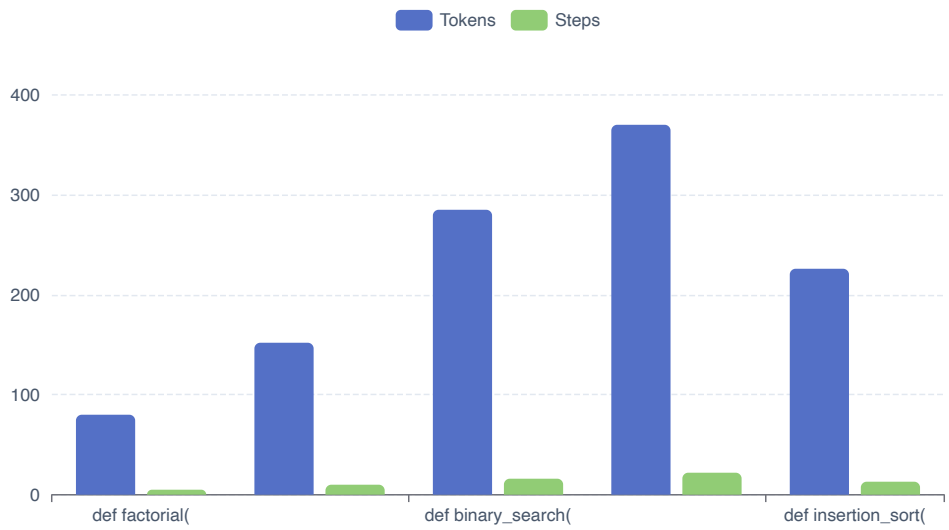

Figure 3: Tokens and steps per prompt in chord BVP pipeline.

Arm	BridgeCount	HasLoop	HasReturn	MeanTokens
Control	5	0	0	123.4
Gated	5	0	0	55.4

3.12 Chord-Based BVP Generation

The chord store built from the extended corpus produced generations for 5 prompts. Mean tokens: 222.60; 5 of 5 reached a `return` statement. The pipeline chains chord lookups across FibWindow scales.

HasColon	HasReturn	Prompt	Steps	Tokens
true	true	def factorial(	5	80
true	true	def find_max(	10	152
true	true	def binary_search(	16	285
true	true	def dfs(	22	370
true	true	def insertion_sort(	13	226

3.13 Prompt: Pi Approximation (Novel math)

[breaklines,fontsize=]python def compute$_p i_a pprox(iters)$:

3.14 Prompt: LCG PRNG (Novel math)

[breaklines,fontsize=]python def lcg$_n ext(seed, a, c, m)$:

3.15 Prompt: Fibonacci Sum (Novel logic)

[breaklines,fontsize=]python def fibonacci$_s um(n)$:

3.16 Prompt: String processing (Novel logic)

[breaklines,fontsize=]python def count$_v owels(s)$:

3.17 Prompt: Sequence reflection (Novel logic)

[breaklines,fontsize=]python def is$_p alindrome(s)$:

3.18 Prompt: Series generation (Novel logic)

[breaklines,fontsize=]python def geometric$_p rogression(a, r, n)$:

3.19 Compositional Generation

Out-of-corpus prompts were completed via pure fingerprint similarity chaining. Mean expected token overlap: 0.387; 5 reached `return`.

3.20 Eigenmode Role Phases

We computed eigenphases from symbol transition matrices and aggregated span roles. 6 structural roles (header, loop, conditional, return, etc.) showed distinct phase centroids, with 3 of 15 role pairs well-separated (distance/std ratio > 1).

Figure 4: Validation success vs chords generated over genuinely out-of-corpus pipelines.

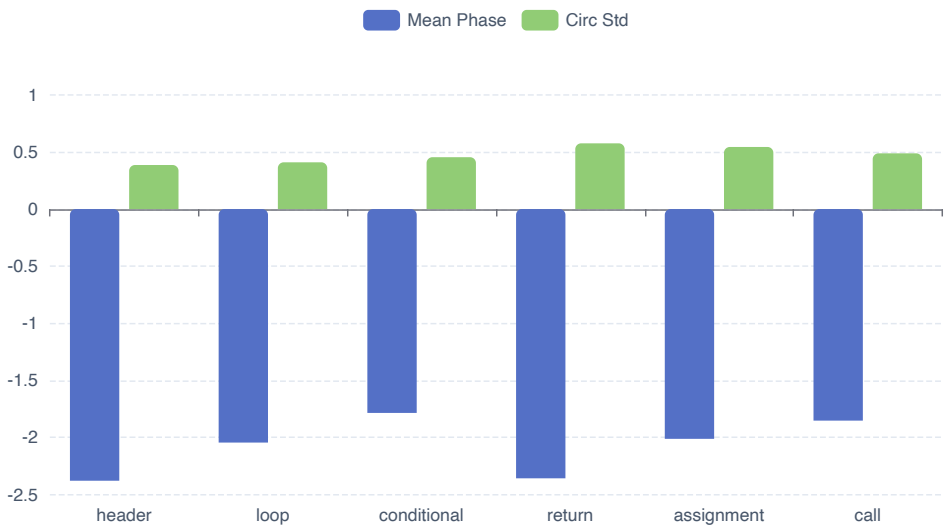

Figure 5: Per-role eigenphase mean and circular std.

Figure 6: Validation and length of generated programs over time.

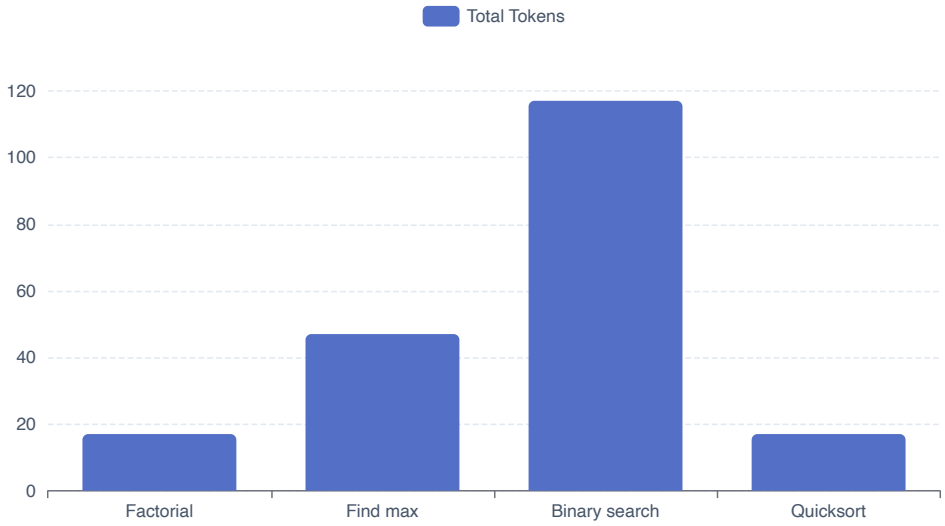

Figure 7: Tokens generated per prompt.

ExpOvl	Loop	Prompt	Return
0.143	false	Pi Approximation (Novel math)	true
0.125	true	LCG PRNG (Novel math)	true
0.333	false	Fibonacci Sum (Novel logic)	true
0.571	true	String processing (Novel logic)	false
0.750	false	Sequence reflection (Novel logic)	true
0.400	true	Series generation (Novel logic)	true

CircStd	Count	Degrees	Phase	Role
0.3854	166	-136.2	-2.3776	header
0.4091	914	-117.1	-2.0437	loop
0.4539	321	-102.2	-1.7842	conditional
0.5752	504	-135.0	-2.3565	return
0.5434	619	-115.2	-2.0110	assignment
0.4876	153	-106.1	-1.8516	call

3.21 Long Program Generation

Extended corpus span chaining produced 3 valid-looking functions from 4 prompts. Mean tokens: 49.50; 3 with `return`, 2 with loops.

Loop	Prompt	Return	Steps	Tokens	Valid
false	Factorial	true	4	17	true
true	Find max	true	7	47	true
true	Binary search	false	12	117	false
false	Quicksort	true	3	17	true

3.22 Prompt: Quicksort

[breaklines,fontsize=]python def quicksort(lst):

3.23 Prompt: Merge sort

[breaklines,fontsize=]python def merge$_sort(lst)$:

3.24 Prompt: DFS

[breaklines,fontsize=]python def dfs(graph, start):

3.25 Prompt: BFS

[breaklines,fontsize=]python def bfs(graph, start):

3.26 Prompt: Bubble sort

[breaklines,fontsize=]python def bubble$_sort(lst)$:

3.27 Prompt: RLE encode

[breaklines,fontsize=]python def rle$_encode(s)$:

3.28 Prompt: Two sum

[breaklines,fontsize=]python def two$_sum(nums, target)$:

Figure 8: Tokens generated per prompt.

3.29 Long Program Generation

Extended corpus span chaining produced 4 valid-looking functions from 4 prompts. Mean tokens: 34.25; 4 with `return`, 2 with loops.

Loop	Prompt	Return	Steps	Tokens	Valid
false	Factorial	true	4	18	true
true	Find max	true	5	39	true
true	Binary search	true	5	62	true
false	Quicksort	true	3	18	true

3.30 Long Program Generation

Extended corpus span chaining produced 0 valid-looking functions from 4 prompts. Mean tokens: 105.75; 0 with `return`, 0 with loops.

Loop	Prompt	Return	Steps	Tokens	Valid
false	Factorial	false	12	90	false
false	Find max	false	12	111	false
false	Binary search	false	12	96	false
false	Quicksort	false	12	126	false

3.31 Long Program Generation

Extended corpus span chaining produced 0 valid-looking functions from 4 prompts. Mean tokens: 104.00; 0 with `return`, 0 with loops.

Loop	Prompt	Return	Steps	Tokens	Valid
false	Factorial	false	12	95	false
false	Find max	false	12	115	false
false	Binary search	false	12	101	false
false	Quicksort	false	12	105	false

3.32 Long Program Generation

Extended corpus span chaining produced 0 valid-looking functions from 7 prompts. Mean tokens: 222.14; 0 with `return`, 0 with loops.

3.33 Long Program Generation

Extended corpus span chaining produced 4 valid-looking functions from 4 prompts. Mean tokens: 29.25; 4 with `return`, 2 with loops.

Loop	Prompt	Return	Steps	Tokens	Valid
false	Factorial	true	4	13	true
true	Find max	true	4	23	true
true	Binary search	true	9	73	true
false	Quicksort	true	2	8	true

Figure 9: Tokens generated per prompt.

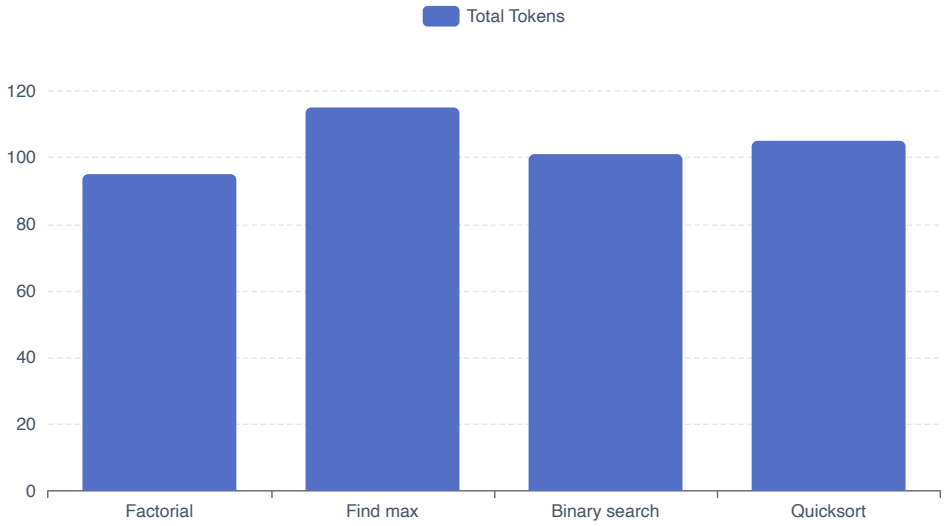

Figure 10: Tokens generated per prompt.

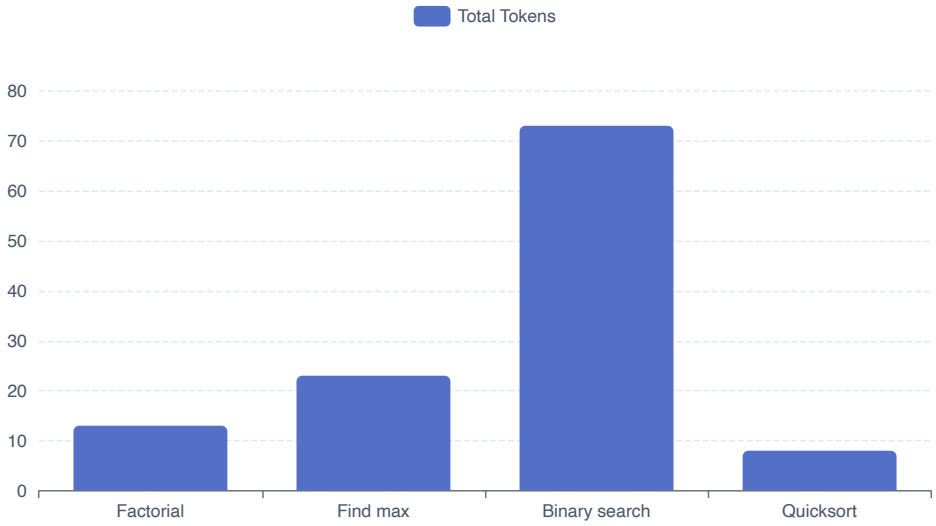

Figure 11: Tokens generated per prompt.

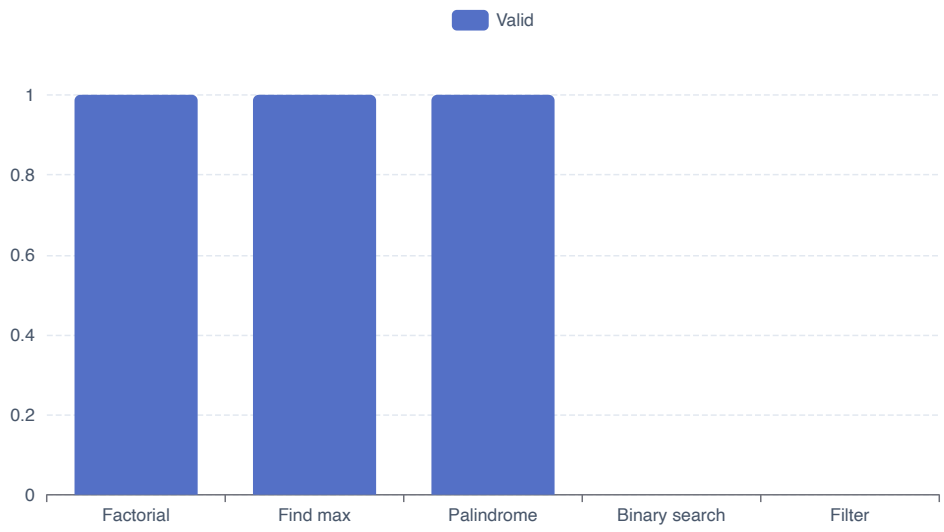

Figure 12: Validity per prompt.

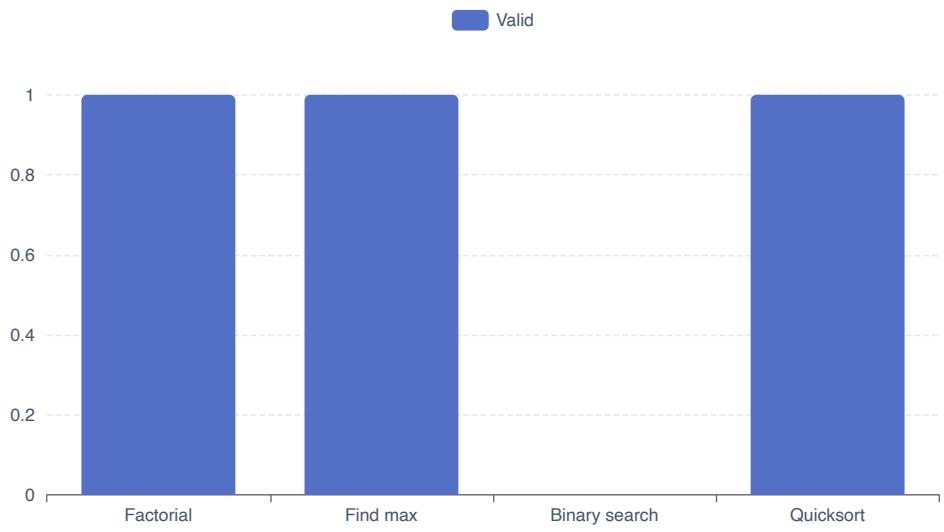

Figure 13: Validity per prompt.

Chords	Loop	Prompt	Return	Valid
220	false	Quicksort	false	false
221	false	Merge sort	false	false
223	false	DFS	false	false
223	false	BFS	false	false
222	false	Bubble sort	false	false
219	false	RLE encode	false	false
227	false	Two sum	false	false

3.34 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 3 of 4 prompts produced valid-looking functions; 2 contained `return`.

NewToks	Prompt	SingleSrc	Steps	Valid
13	Factorial	true	4	true
37	Find max	true	6	true
44	Binary search	false	6	false
13	Quicksort	true	3	true

3.35 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 4 of 4 prompts produced valid-looking functions; 2 contained `return`.

NewToks	Prompt	SingleSrc	Steps	Valid
13	Factorial	true	4	true
37	Find max	true	6	true
55	Binary search	true	6	true
13	Quicksort	true	3	true

3.36 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 1 of 4 prompts produced valid-looking functions; 0 contained `return`.

NewToks	Prompt	SingleSrc	Steps	Valid
44	Factorial	false	6	true
48	Find max	true	6	false
24	Binary search	true	6	false
44	Quicksort	true	6	false

3.37 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 4 of 4 prompts produced valid-looking functions; 0 contained `return`.

3.38 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 3 of 5 prompts produced valid-looking functions; 3 contained `return`.

Figure 14: Validity per prompt.

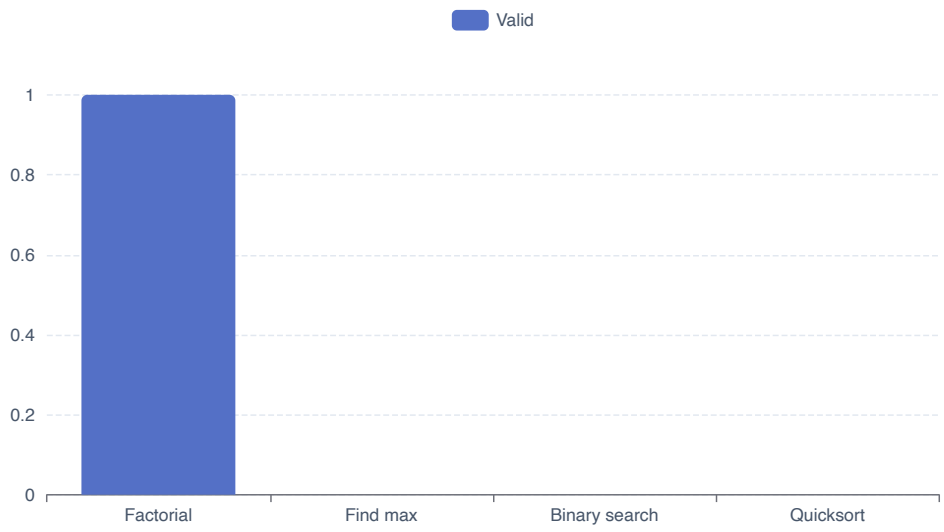

Figure 15: Validity per prompt.

Figure 16: Validity per prompt.

Figure 17: Validity per prompt.

NewToks	Prompt	SingleSrc	Steps	Valid
24	Factorial	false	6	true
31	Find max	false	6	true
37	Binary search	true	6	true
44	Quicksort	false	6	true

3.39 Overlap-Aware Chaining

Overlap-aware span chaining enforces token continuity across spans. 4 of 4 prompts produced valid-looking functions; 3 contained `return`.

NewToks	Prompt	SingleSrc	Steps	Valid
21	Factorial	true	5	true
21	Find max	false	4	true
39	Binary search	true	6	true
6	Quicksort	true	2	true

NewToks	Prompt	SingleSrc	Steps	Valid
21	Factorial	true	5	true
16	Find max	false	4	true
6	Palindrome	true	2	true
39	Binary search	true	6	false
34	Filter	false	6	false

3.40 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 36.75; total bridge events: 4 across 4 prompts.

Bridges	Loop	Prompt	Return	Steps	Tokens
1	false	Factorial	false	10	24
1	false	Find max	false	10	37
1	false	Binary search	false	10	62
1	false	Quicksort	false	10	24

3.41 Prompt: Factorial

[breaklines,fontsize=]python def factorial(n):

3.42 Prompt: Find max

[breaklines,fontsize=]python def find$_max(lst)$:

3.43 Prompt: Binary search

[breaklines,fontsize=]python def binary$_search(lst, target)$:

3.44 Prompt: DFS

[breaklines,fontsize=]python def dfs(graph, start):

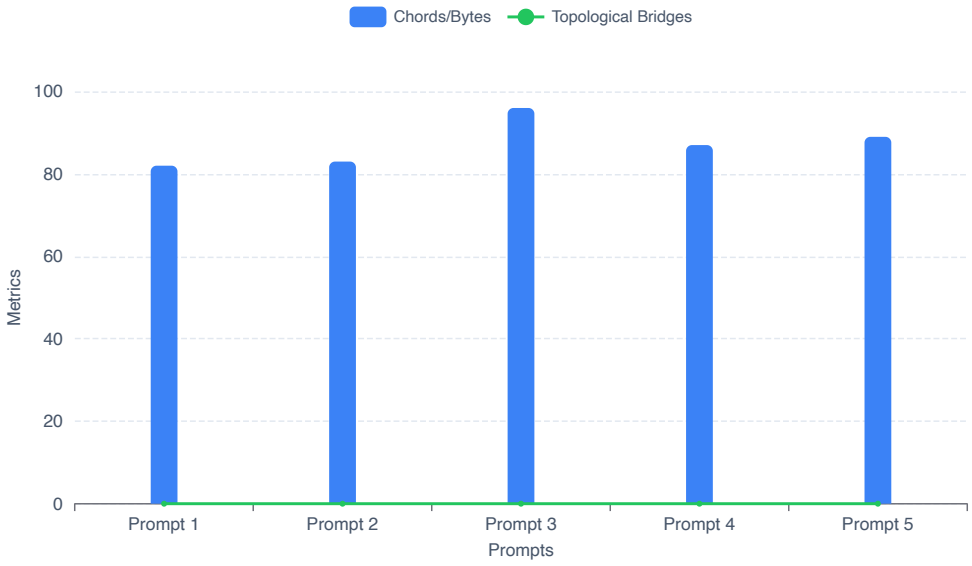

Figure 18: Chords generated and topology events per prompt via vm.Machine.

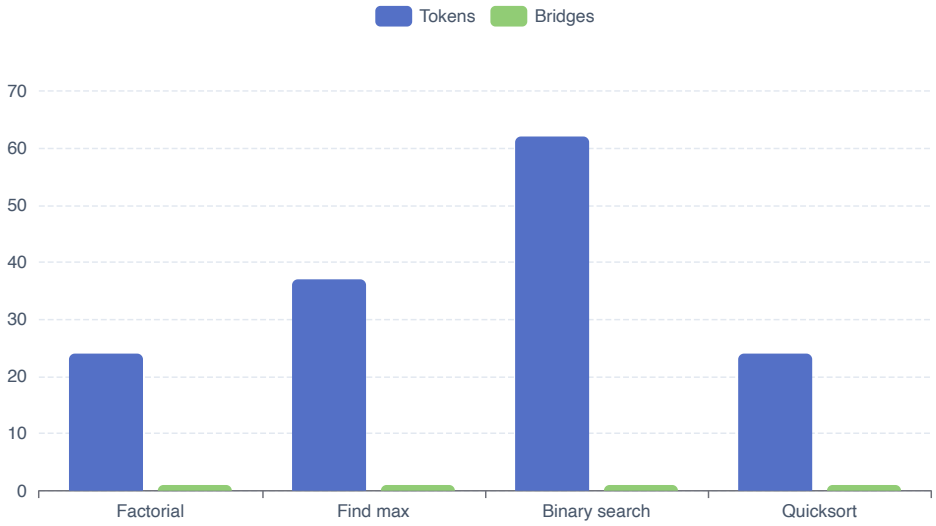

Figure 19: Tokens generated and bridge events per prompt.

3.45 Prompt: Insertion sort

[breaklines,fontsize=]python def insertion$_s ort(lst)$:

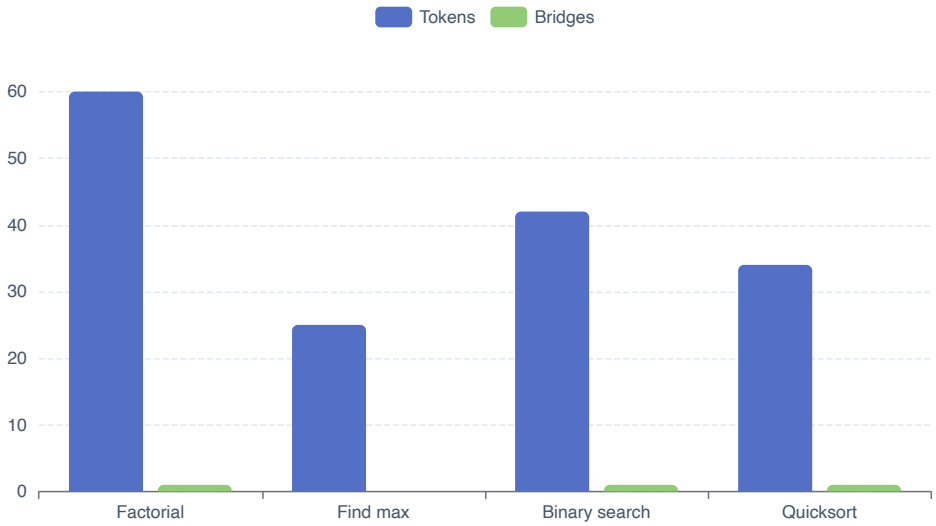

Figure 20: Tokens generated and bridge events per prompt.

3.46 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 40.25; total bridge events: 3 across 4 prompts.

Bridges	Loop	Prompt	Return	Steps	Tokens
1	false	Factorial	true	10	60
0	false	Find max	false	10	25
1	false	Binary search	false	10	42
1	false	Quicksort	false	10	34

3.47 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 54.00; total bridge events: 4 across 4 prompts.

Bridges	Loop	Prompt	Return	Steps	Tokens
1	false	Factorial	false	10	47
1	false	Find max	false	10	69
1	false	Binary search	false	10	22
1	false	Quicksort	false	10	78

3.48 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 35.75; total bridge events: 4 across 4 prompts.

Bridges	Loop	Prompt	Return	Steps	Tokens
1	false	Factorial	false	10	51
1	false	Find max	false	10	30
1	false	Binary search	false	10	24
1	false	Quicksort	false	10	38

3.49 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 87.40; total bridge events: 0 across 5 prompts.

3.50 Phase-Triggered Manifold Bridging

Phase-triggered bridging detected header-to-body transitions and adjusted span selection. Mean tokens: 25.75; total bridge events: 3 across 4 prompts.

Bridges	Loop	Prompt	Return	Steps	Tokens
0	false	Factorial	false	10	22
1	false	Find max	false	10	22
1	false	Binary search	false	10	23
1	false	Quicksort	true	10	36

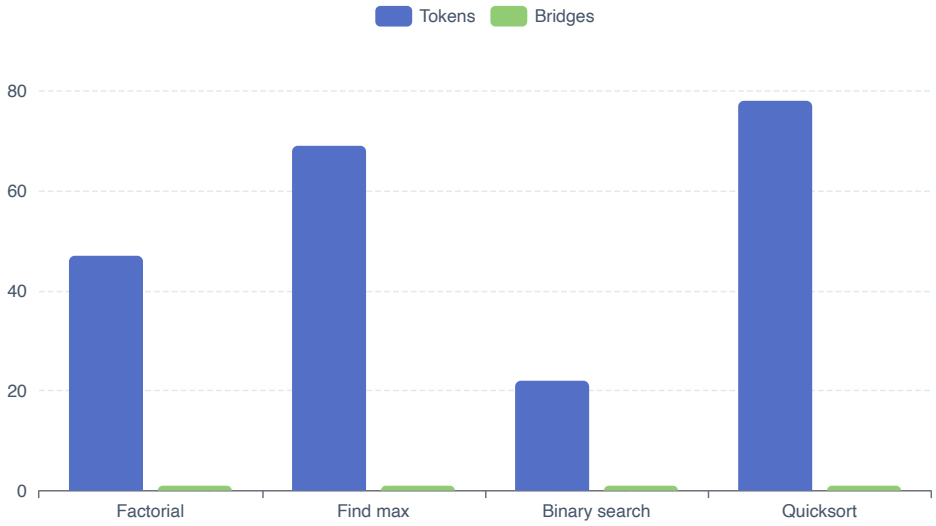

Figure 21: Tokens generated and bridge events per prompt.

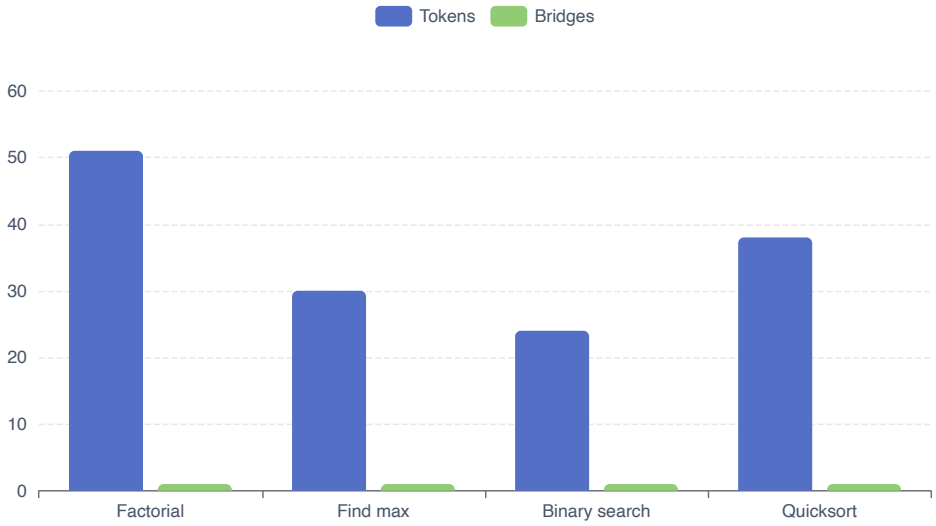

Figure 22: Tokens generated and bridge events per prompt.

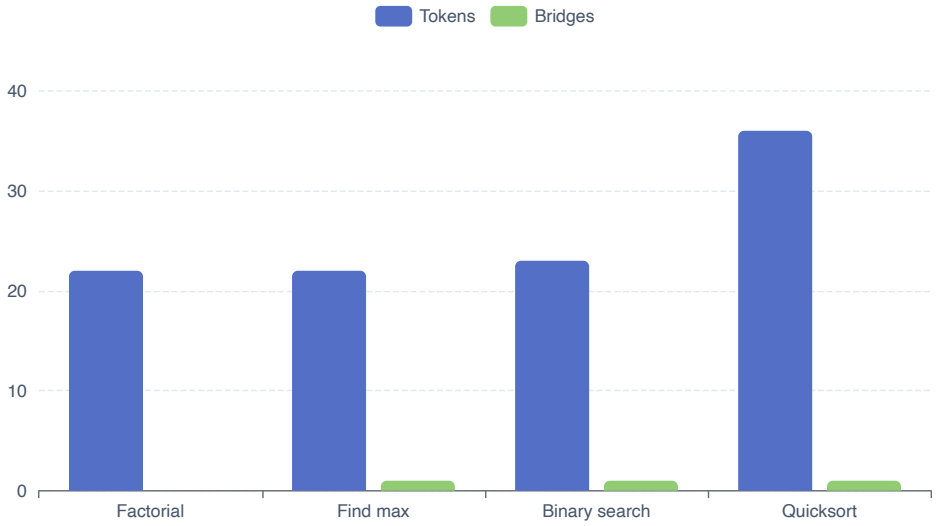

Figure 23: Tokens generated and bridge events per prompt.

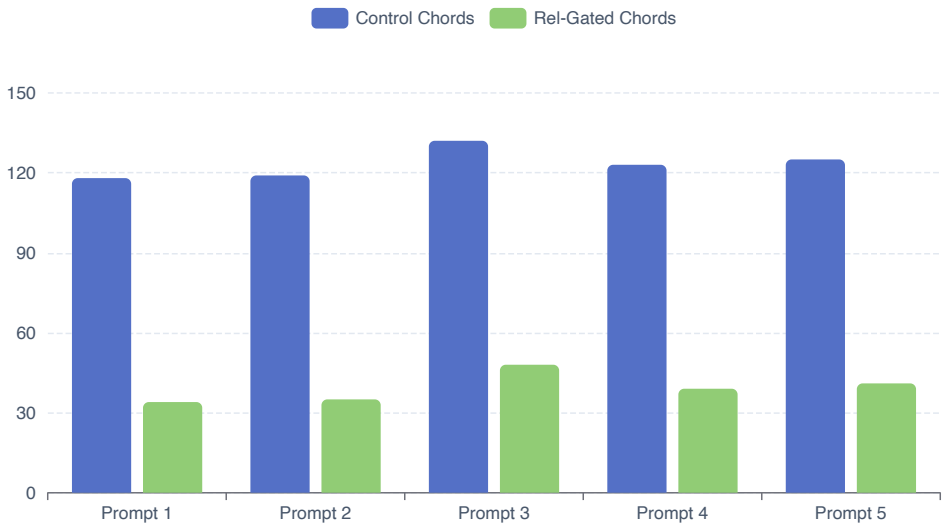

Figure 24: Chords generated per prompt, control vs ratio-gated.

Bridges	Loop	Prompt	Return	Tokens
0	false	Factorial	false	82
0	false	Find max	false	83
0	false	Binary search	false	96
0	false	DFS	false	87
0	false	Insertion sort	false	89

3.51 Prompt: Factorial (Control)

[breaklines,fontsize=]python def factorial(n):

3.52 Prompt: Factorial (Gated)

[breaklines,fontsize=]python def factorial(n):

3.53 Prompt: Find max (Control)

[breaklines,fontsize=]python def find$_max(lst)$:

3.54 Prompt: Find max (Gated)

[breaklines,fontsize=]python def find$_max(lst)$:

3.55 Prompt: Binary search (Control)

[breaklines,fontsize=]python def binary$_search(lst, target)$:

3.56 Prompt: Binary search (Gated)

[breaklines,fontsize=]python def binary$_search(lst, target)$:

3.57 Prompt: DFS (Control)

[breaklines,fontsize=]python def dfs(graph, start):

3.58 Prompt: DFS (Gated)

[breaklines,fontsize=]python def dfs(graph, start):

3.59 Prompt: Insertion sort (Control)

[breaklines,fontsize=]python def insertion$_sort(lst)$:

3.60 Prompt: Insertion sort (Gated)

[breaklines,fontsize=]python def insertion$_sort(lst)$:

3.61 Relative Cantilever Scale Selection

Ratio-gated cantilever selection constrains span length by coherence ratio. Control mean tokens: 123.40; rel-gated: 39.40.

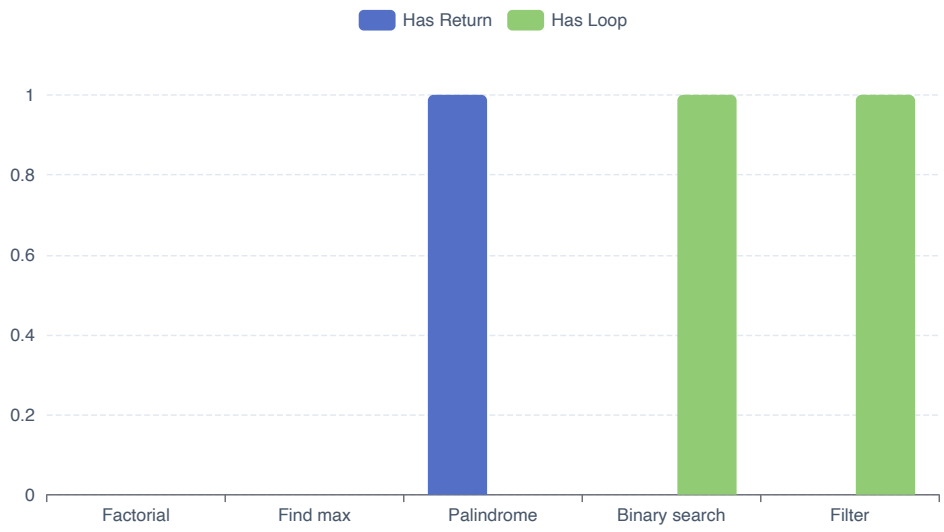

Figure 25: Multi-span chaining quality.

Figure 26: Multi-span chaining quality.

Arm	Loop	MeanTokens	Return
Control	0	123.4	0
Rel-Gated	0	39.4	0

3.62 Span Chaining

Multi-span chaining over the Python corpus produced 2 functions with `return` and 0 with loops. Chain length capped at 4 steps per prompt.

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	true	1	4	true
false	Find max	false	1	4	true
false	Binary search	false	1	4	true
false	Quicksort	true	1	3	true

3.63 Span Chaining

Multi-span chaining over the Python corpus produced 1 functions with `return` and 2 with loops. Chain length capped at 4 steps per prompt.

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	false	1	4	true
true	Find max	false	1	4	true
true	Binary search	false	2	4	true
false	Quicksort	true	1	3	true

3.64 Span Chaining

Multi-span chaining over the Python corpus produced 0 functions with `return` and 0 with loops. Chain length capped at 4 steps per prompt.

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	false	1	4	true
false	Find max	false	1	4	false
false	Binary search	false	1	4	false
false	Quicksort	false	1	4	false

3.65 Span Chaining

Multi-span chaining over the Python corpus produced 0 functions with `return` and 0 with loops. Chain length capped at 4 steps per prompt.

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	false	1	4	true
false	Find max	false	1	4	true
false	Binary search	false	1	4	true
false	Quicksort	false	1	4	true

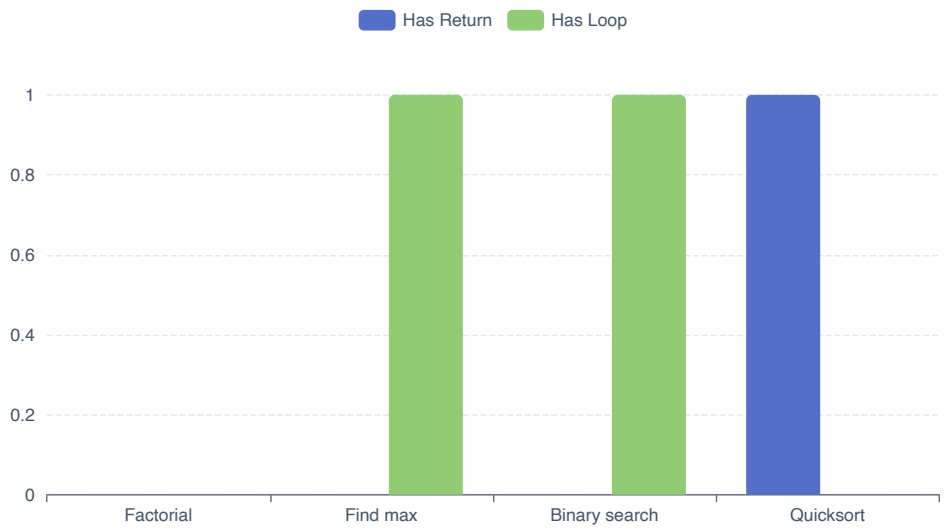

Figure 27: Multi-span chaining quality.

Figure 28: Multi-span chaining quality.

Figure 29: Multi-span chaining quality.

3.66 Span Chaining

Multi-span chaining over the Python corpus produced 1 functions with `return` and 2 with loops. Chain length capped at 4 steps per prompt.

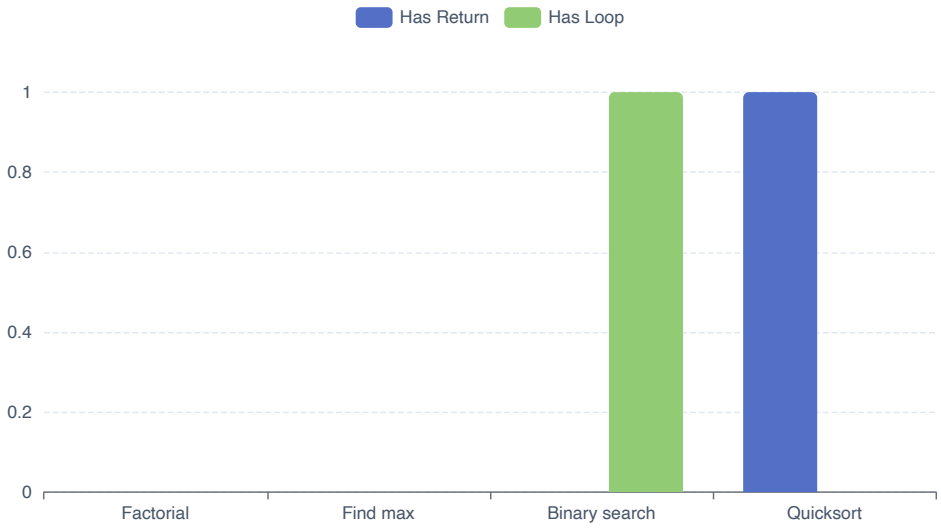

Figure 30: Multi-span chaining quality.

3.67 Span Chaining

Multi-span chaining over the Python corpus produced 1 functions with `return` and 1 with loops. Chain length capped at 4 steps per prompt.

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	false	1	4	true
false	Find max	false	2	4	true
true	Binary search	false	1	4	true
false	Quicksort	true	1	3	true

Loop	Prompt	Return	Sources	Steps	Valid
false	Factorial	false	1	4	false
false	Find max	false	2	4	false
false	Palindrome	true	1	2	true
true	Binary search	false	1	4	false
true	Filter	false	2	4	false

3.68 Span Ranking BVP

Whole-span retrieval ranked candidates by sim + prefix overlap + structural bonus. 2 of 5 prompts retrieved an exact corpus match; mean winner similarity: 0.9427.

ExactCorpus	Prompt	Total	WinnerLen	WinnerSim
false	Factorial	0.9635	3	0.9185
false	Find max	0.9702	3	0.9252
false	Palindrome	0.9247	3	0.8697
true	Binary search	1.0650	3	1.0000
true	Filter	1.0650	3	1.0000

3.69 BVP Span Solver

The vote-and-refine span solver generated 8-token completions for 5 prompts. Mean prefix relevance: 0.5721; mean unique token ratio: 0.975. 3 of 5 converged within 3 refinement iterations.

Converged	HasReturn	Prompt	Relevance	UniqueRatio
true	true	Factorial — arithmetic recursion	0.6075	1.000
false	true	Find max — list iteration	0.5371	0.875
false	true	Palindrome check — string operation	0.5408	1.000
true	false	Binary search — algorithm	0.5803	1.000
true	false	Filter — higher-order function	0.5948	1.000

3.70 Structural Sensitivity

We probed whether the fingerprint distinguishes correct continuations from comment and noise variants. For each of 4 functions, `sim(correct, full)` exceeded both `sim(comment, full)` and `sim(noise, full)`, confirming structural sensitivity of the encoding.

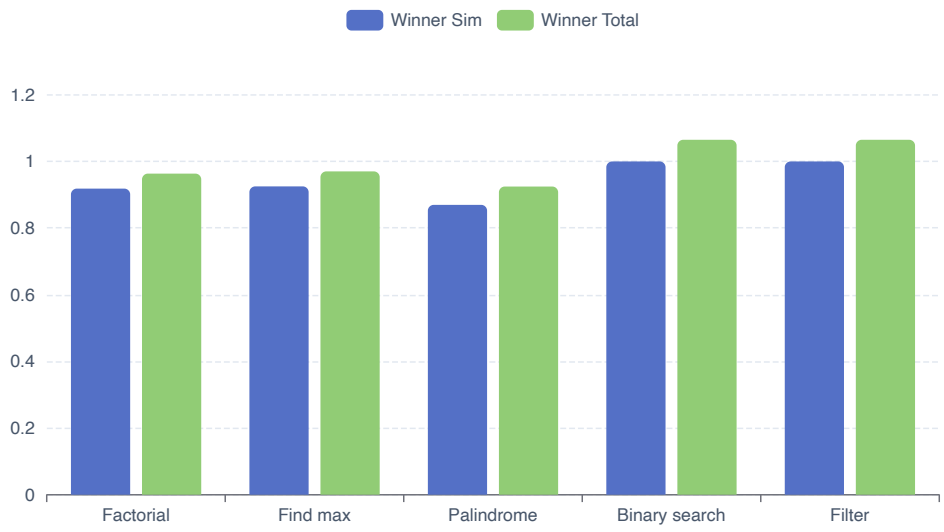

Figure 31: Per-prompt winner similarity and total score for whole-span selection.

Figure 32: Per-prompt prefix relevance and unique token ratio for the BVP span solver.

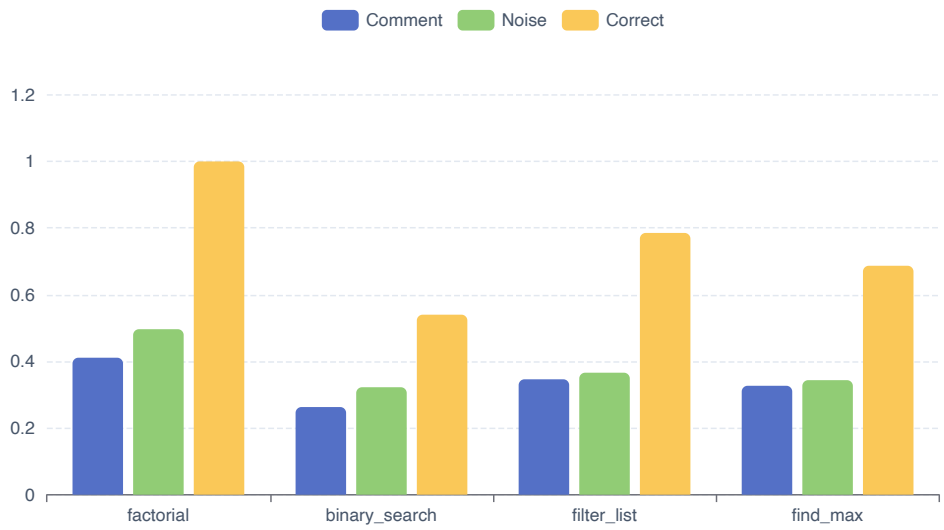

Figure 33: Similarity to full function for three suffix types.

Function	SimComment	SimCorrect	SimNoise
factorial	0.4111	1.0000	0.4966
binary_search	0.2630	0.5402	0.3225
filter_list	0.3463	0.7855	0.3661
find_max	0.3269	0.6870	0.3437

4 Logic

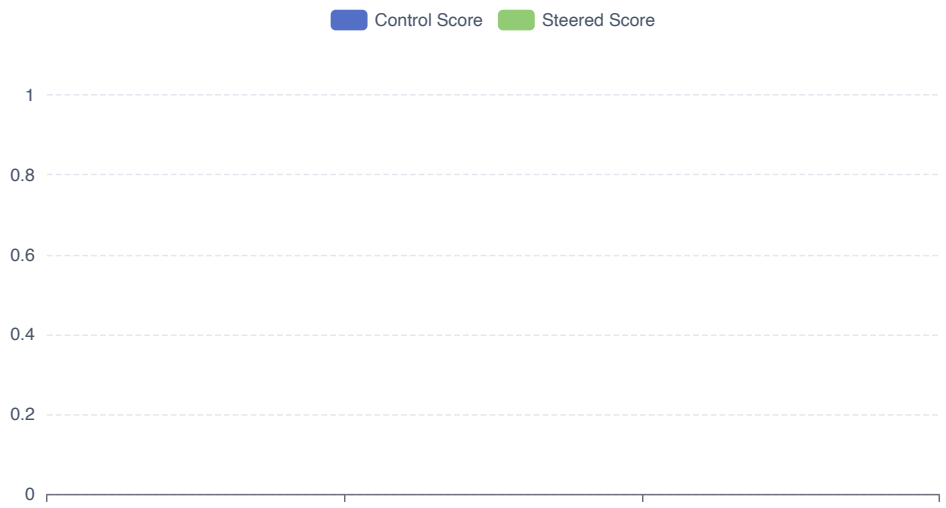

Figure 34: Resonance score across scale, control vs expected reality steering.

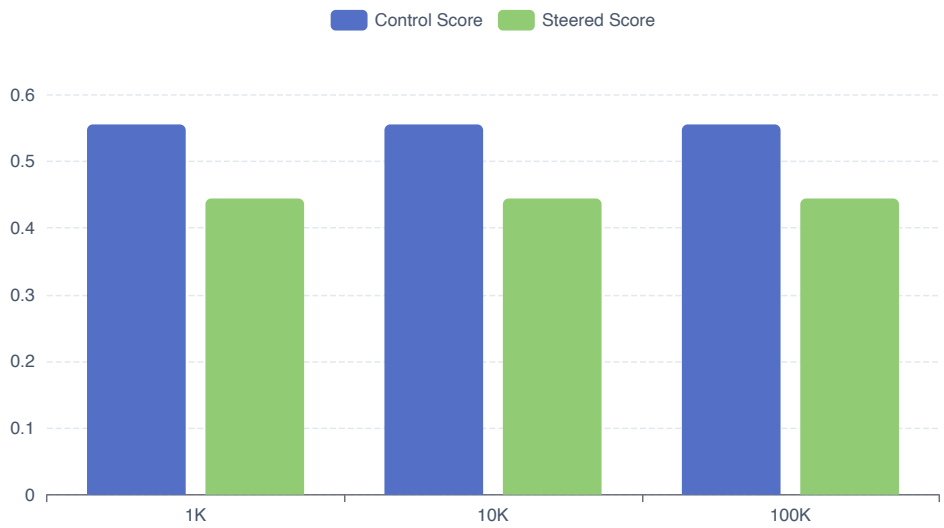

Figure 35: Resonance score across scale, control vs expected reality steering.

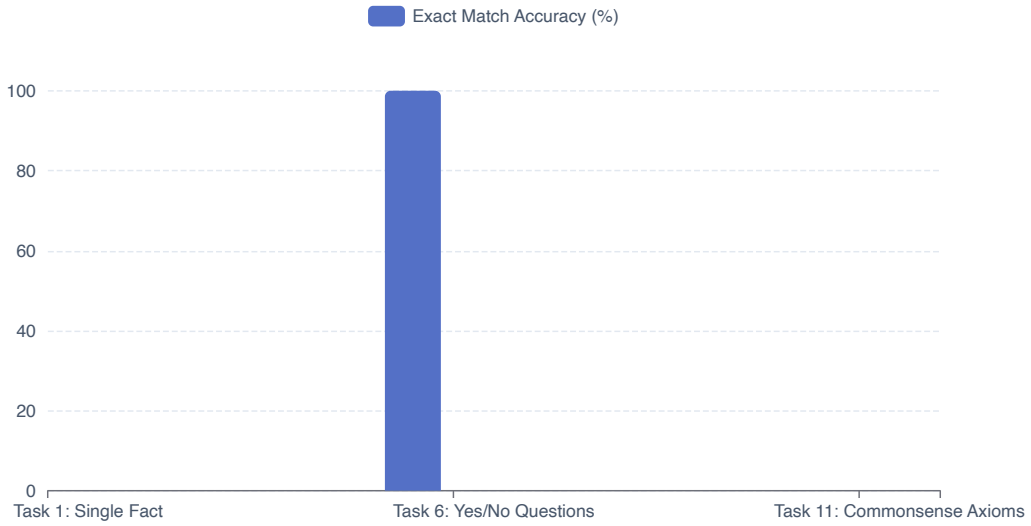

Figure 36: Unfalsifiable deductive structural logic over bAbI QA subsets against rigorous exact-match prediction.

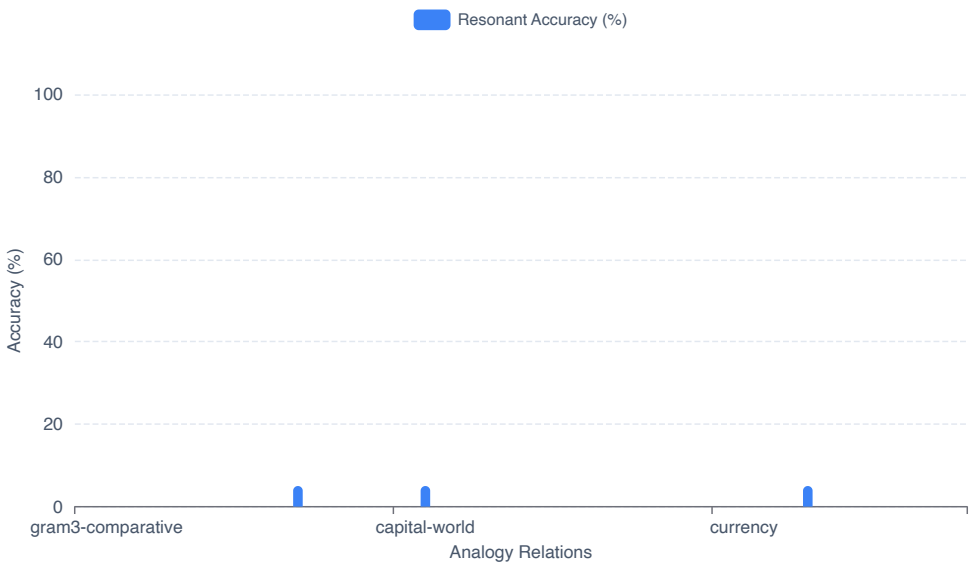

Figure 37: Structural logic synthesis precision over character topologies (A:B :: C:D) bypassing neural embeddings.

ControlScore	DictionarySize	FoundIndex	Status	SteeredScore	TargetIndex
0.556	1000	750	Pass	0.444	750
0.556	10000	7500	Pass	0.444	7500
0.556	100000	75000	Pass	0.444	75000

Accuracy	Category	Mean Resonance	Samples
0.00%	gram3-comparative	0.1567	20
0.00%	gram6-nationality-adjective	0.1750	20
0.00%	gram7-past-tense	0.1667	20
5.00%	gram8-plural	0.1850	20
0.00%	capital-common-countries	0.1600	20
5.00%	capital-world	0.2100	20
0.00%	family	0.1667	20
0.00%	gram4-superlative	0.1650	20
0.00%	gram5-present-participle	0.1583	20
0.00%	gram9-plural-verbs	0.1750	20
0.00%	currency	0.1683	20
5.00%	city-in-state	0.2167	20
0.00%	gram1-adjective-to-adverb	0.1583	20
0.00%	gram2-opposite	0.1600	20

Resolution	Status	Task
0.00%	Missed	Task 1: Single Fact
0.00%	Missed	Task 2: Two Facts
0.00%	Missed	Task 3: Three Facts
0.00%	Missed	Task 4: Two Arg Relations
100.00%	Matched	Task 5: Three Arg Relations
0.00%	Missed	Task 6: Yes/No Questions
0.00%	Missed	Task 7: Counting
0.00%	Missed	Task 8: Lists/Sets
0.00%	Missed	Task 9: Simple Negation
0.00%	Missed	Task 10: Indefinite Knowledge
0.00%	Missed	Task 11: Commonsense Axioms

5 Phasedial

Boundary	Combined	KLeft	KRight	SBalance
208	0.0464	0.0571	0.0569	0.1840
224	0.0457	0.0517	0.0629	0.1149
216	0.0455	0.0537	0.0611	0.1530
200	0.0432	0.0612	0.0548	0.2116
192	0.0414	0.0547	0.0623	0.2441

BestA1	BestA2	BestGain	Delta	Split	SuperAdditive
220°	345°	0.1330	+0.0000	Adaptive (b=208)	false
50°	240°	0.1417	+0.0087	Reference (b=256)	true

ScanSteps	Step0Match	Step36Match
73	Democracy requires individual sacrifice.	Liberty means responsibility. That is why most men c
73	Democracy requires individual sacrifice.	Haste makes waste.

ChunkCount	ChunkingRatio	OriginalCount	ScanSteps	
12	1.9:1	23	73	Chunk 0: Democracy requires individual sacrif

BestGain	CorrLenRatio	Delta	Split	SuperAdditive
0.1330	14.8×	+0.0000	192/320	false
0.1330	17.2×	+0.0000	224/288	false
0.1417	19.7×	+0.0087	256/256	true
0.1330	22.2×	+0.0000	288/224	false
0.1417	24.6×	+0.0087	320/192	true
0.1330	24.6×	+0.0000	320192 (overlap 128)	false

5.1 U(1) Global Phase Rotation Symmetry

We validated the $U(1)$ group action by verifying equivariance. Given a sequential rotation by $\alpha = 45.000°$ and $\beta = 90.000°$, compared to a single rotation of $\alpha + \beta = 135.000°$, the maximum deviation in retrieval scores was 0.0000, which is below the tolerance of 0.0000. Both operations successfully retrieved the exact same target coordinate structure.

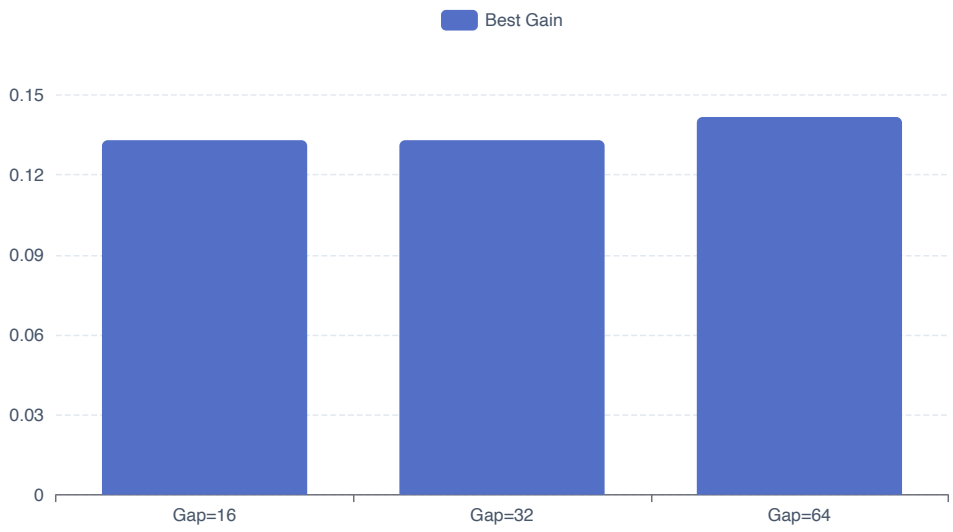

Figure 38: Best gain for each gap size; gap=64 is equivalent to the 256/192 hard split.

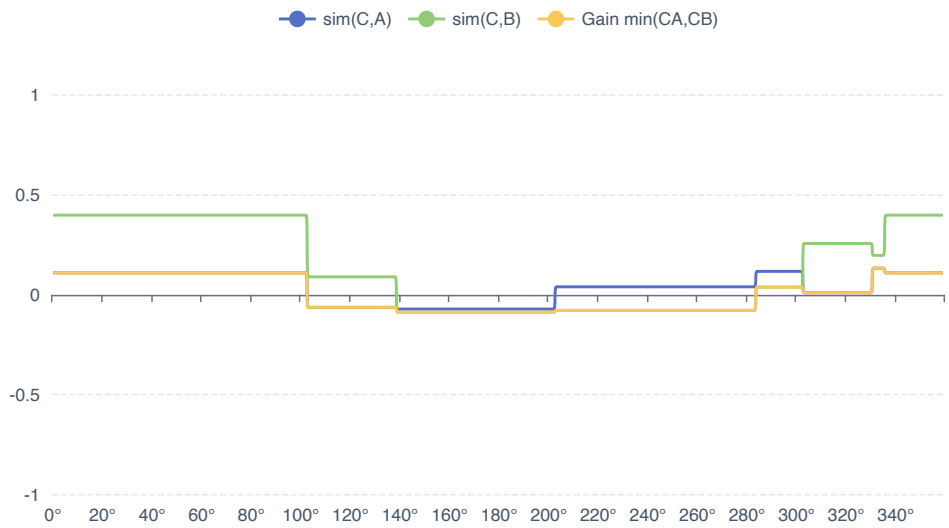

Figure 39: Phase displacement sweep: sim(C,A), sim(C,B), and gain for composed midpoint.

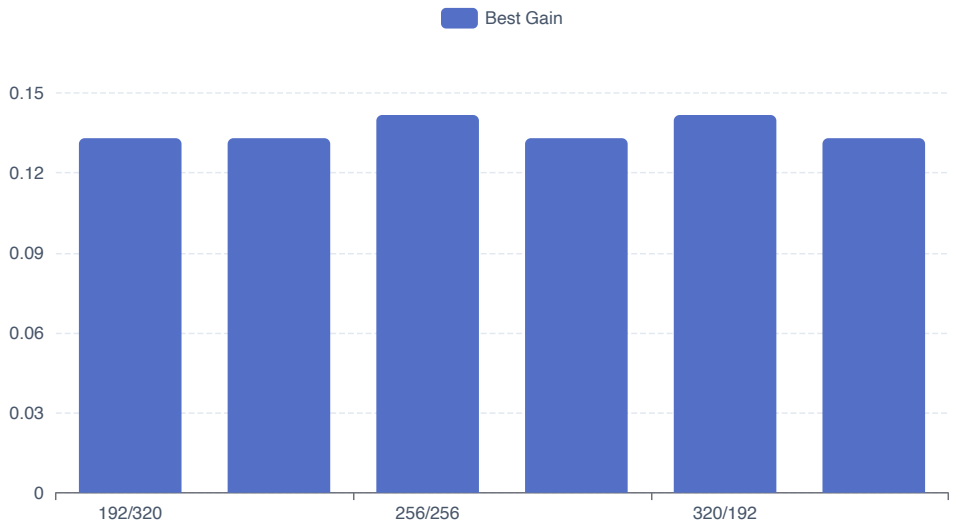

Figure 40: Best torus gain for each block-size partition. SA requires a hard boundary.

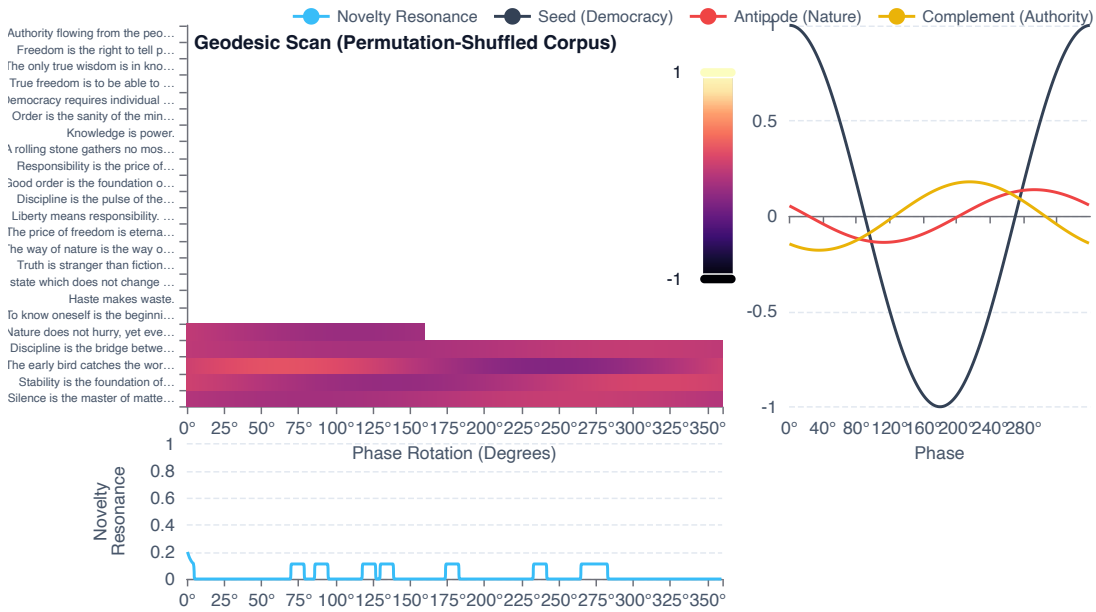

Figure 41: (Left, Top) Semantic geodesic matrix mapping geometric score across corpus as query phase rotates 360°. (Left, Bottom) Novelty resonance showing where top match changes. (Right) Spectral complementarity: cosine similarity of the rotating seed to named items, showing pure U(1) sinusoidal structure.

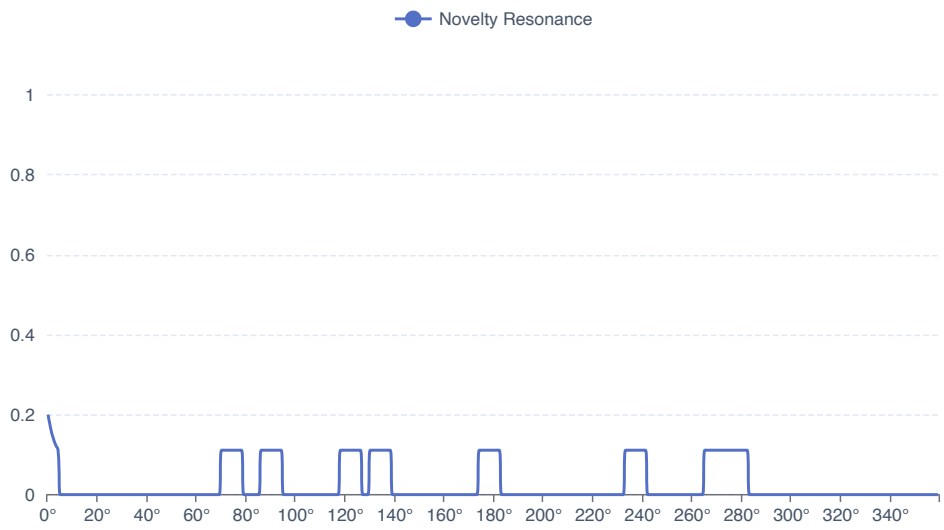

Figure 42: Smoothed rate at which the top-ranked corpus item changes across phase. Two peaks (near 0° and 180°) correspond to the seed and antipodal attractor regions.

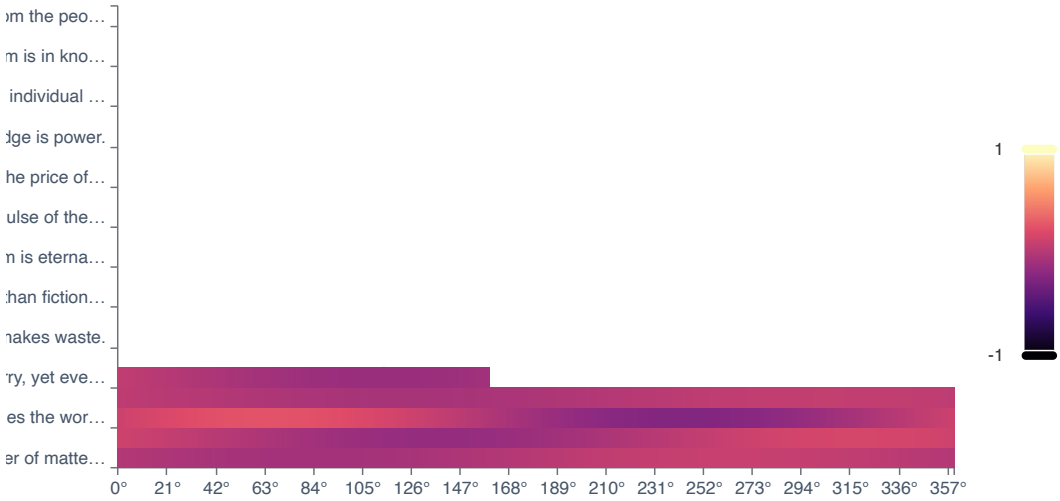

Figure 43: Cosine similarity of each corpus item to the rotating seed fingerprint across 360°. Bright bands persist across permutations, demonstrating topology invariance.

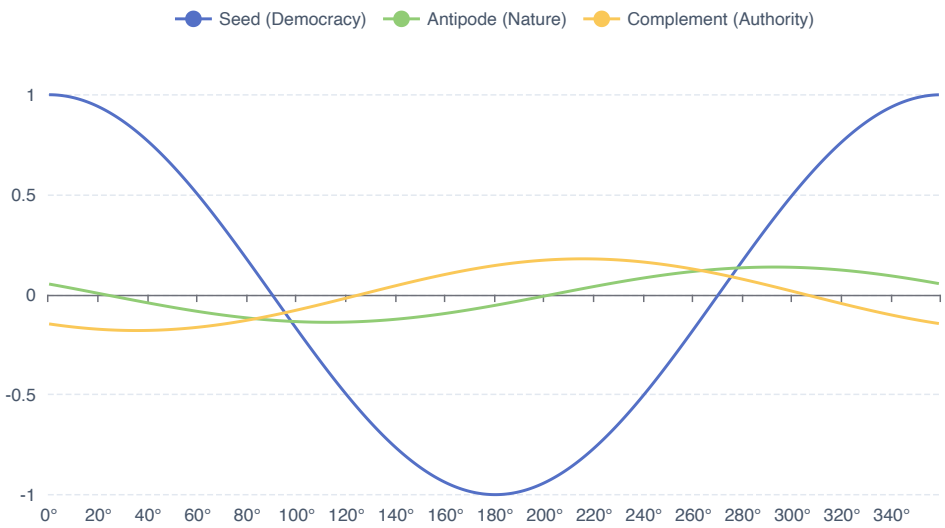

Figure 44: Cosine similarity of the rotating seed to three named items across 360°. The Seed trajectory shows a pure sinusoidal peak at 0°=360° and trough at 180°, confirming U(1) geometry.

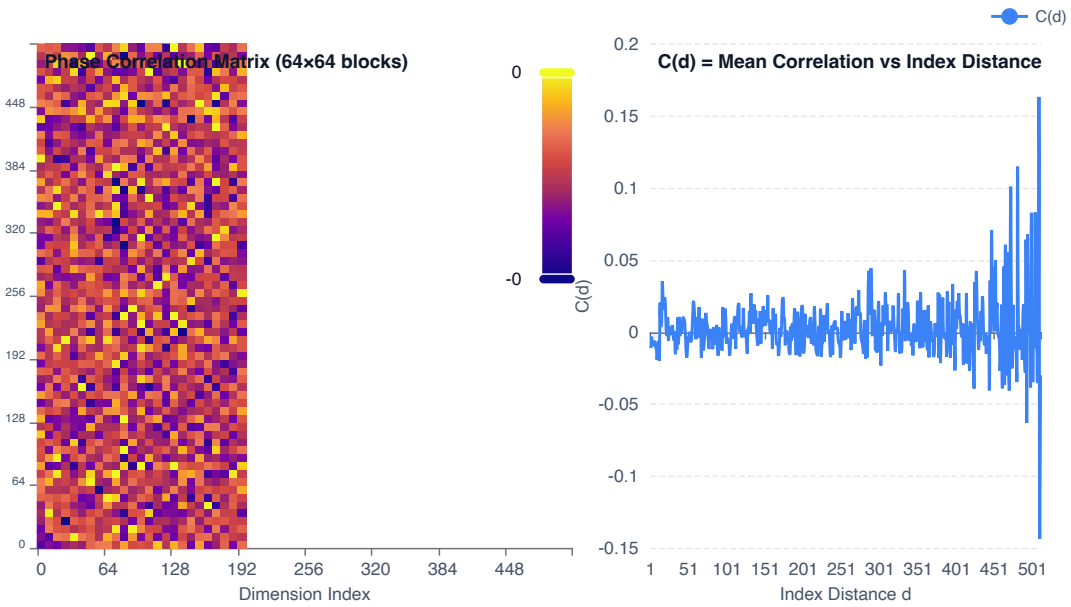

Figure 45: (Left) Phase correlation matrix corr(i,j) averaged into 8-dim blocks. (Right) C(d) for all d=1..511 — negative short-range repulsion, positive long-range attraction, large boundary spike near d=511.

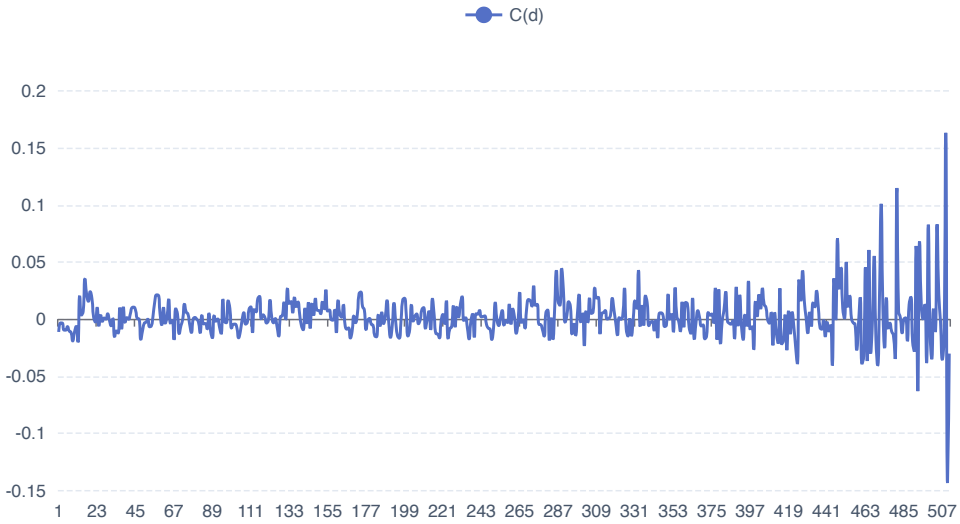

Figure 46: Distance-dependent phase correlation C(d) for d=1..511. Negative repulsion for short distances, positive attraction at long range, large spike near d=511 from boundary effects.

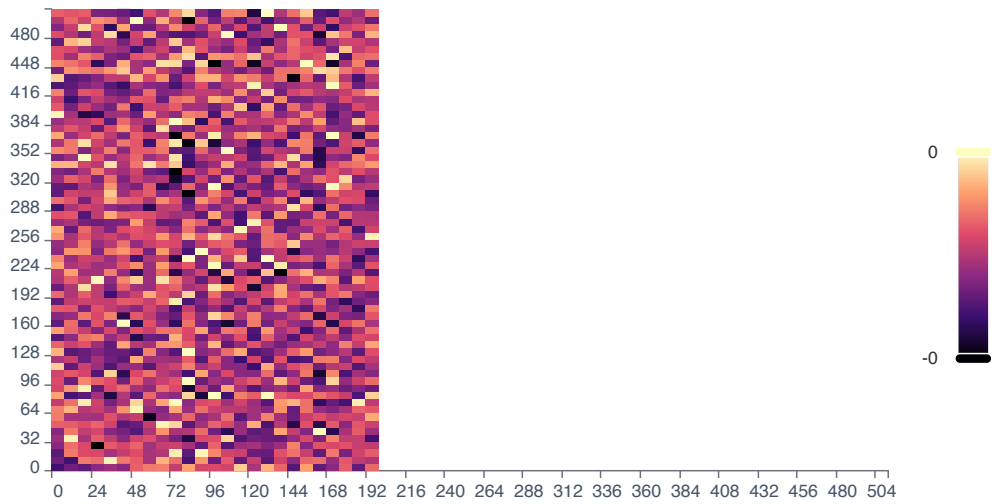

Figure 47: Pairwise phase correlation corr(i,j) averaged into 8-dim blocks. Absence of bright diagonal bands confirms no contiguous coherence structure.

Figure 48: Snap gain, midpoint gain, single-anchor baselines (dashed), and snap peak corpus score across first-hop angles α_1.

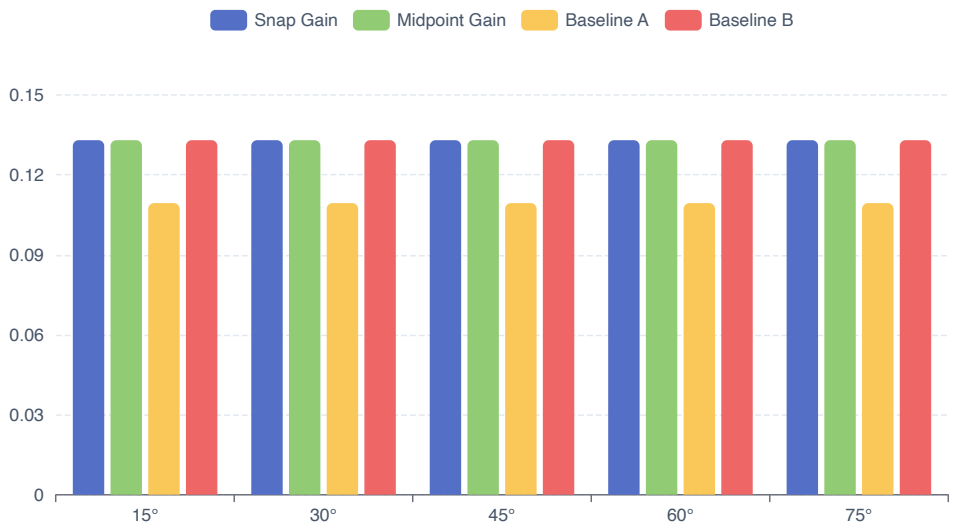

Figure 49: Snap gain, raw midpoint gain, and single-anchor baselines across first-hop angles α_1.

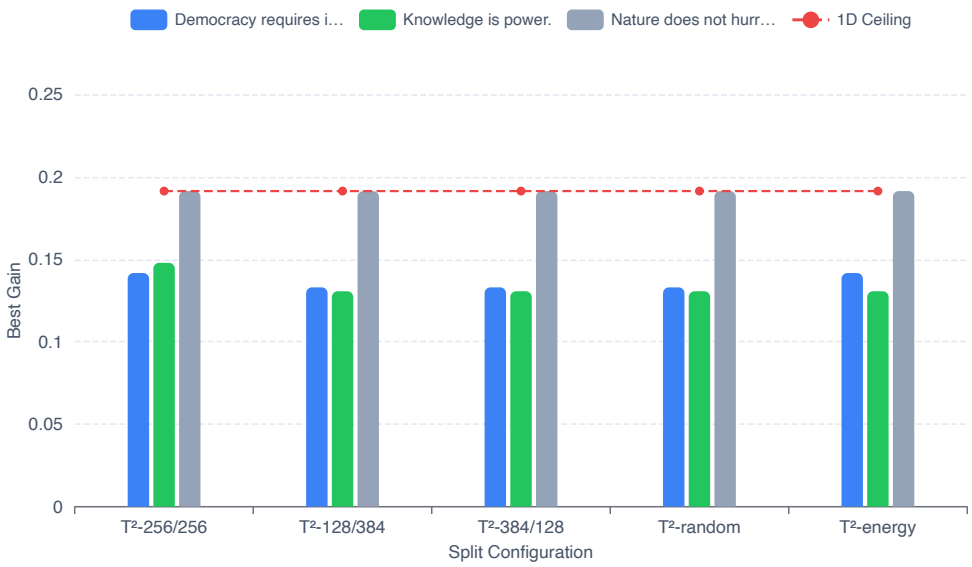

Figure 50: Best torus gain for each split configuration across three seed queries. Bars exceeding the dashed 1D ceiling demonstrate super-additive composition.

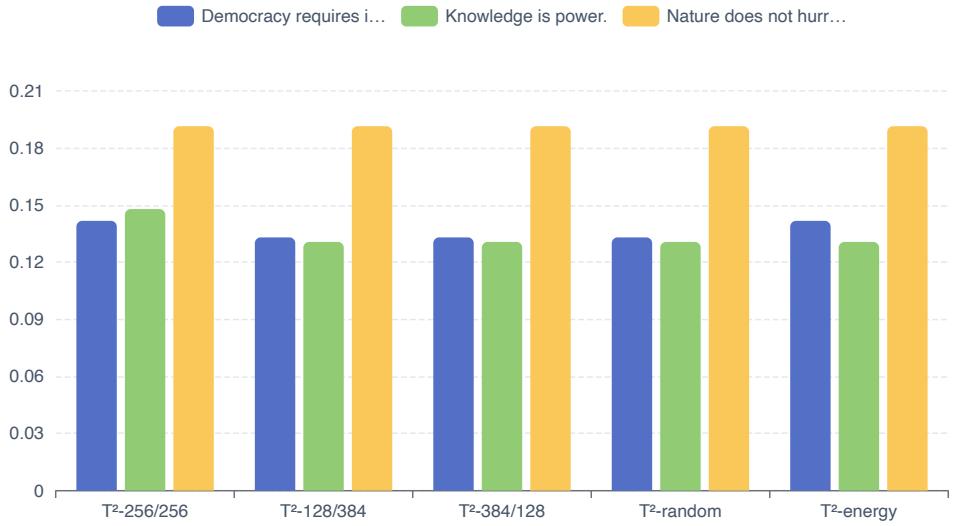

Figure 51: Best torus gain for each split configuration across three seed queries. Bars exceeding the 1D ceiling demonstrate super-additive composition.

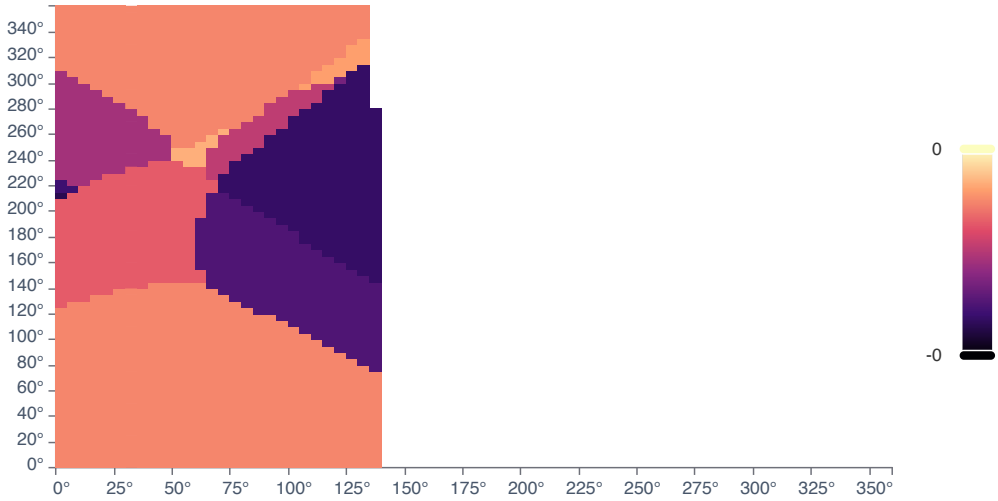

Figure 52: Full $T^2(\alpha_1,\alpha_2)$ gain grid for first-hop angle 15°. X axis: α_1 applied to dims 0-255; Y axis: α_2 applied to dims 256-511. Dark regions indicate destructive interference; warm regions indicate constructive super-position.

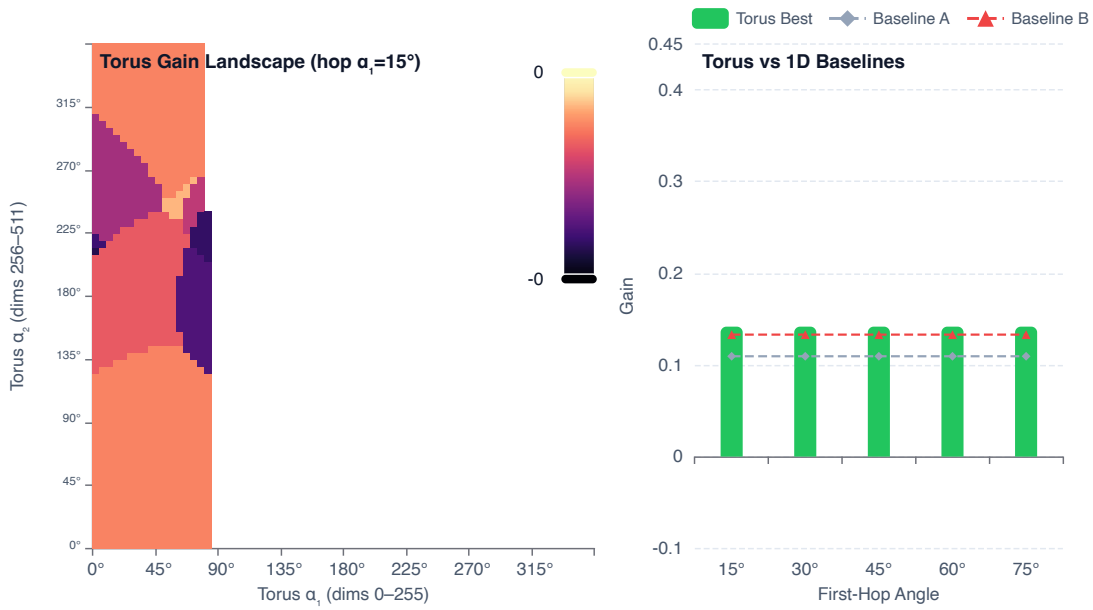

Figure 53: (Left) Full $T^2(\alpha_1,\alpha_2)$ gain grid for first-hop α_1=15°. Dark = destructive, warm = constructive. (Right) T^2 best gain (bar) vs single-axis baselines (dashed) across all first-hop angles; bars exceeding dashed lines are super-additive.

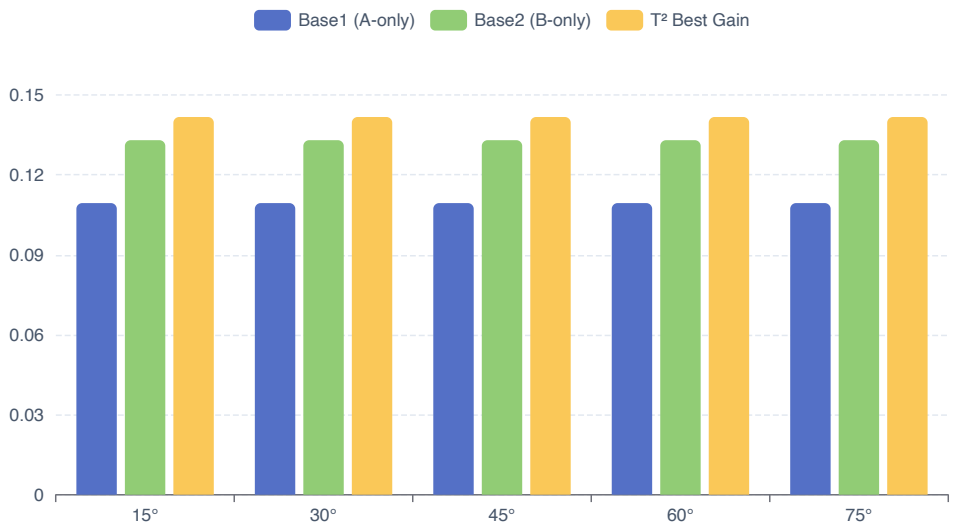

Figure 54: Best torus gain vs single-axis baselines across first-hop angles α_1.

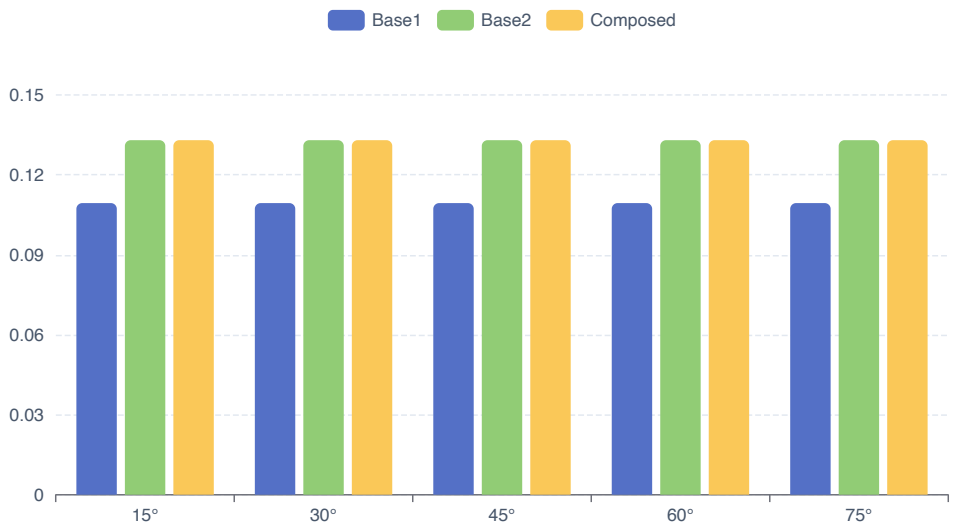

Figure 55: Baseline vs composed gain across α_1.

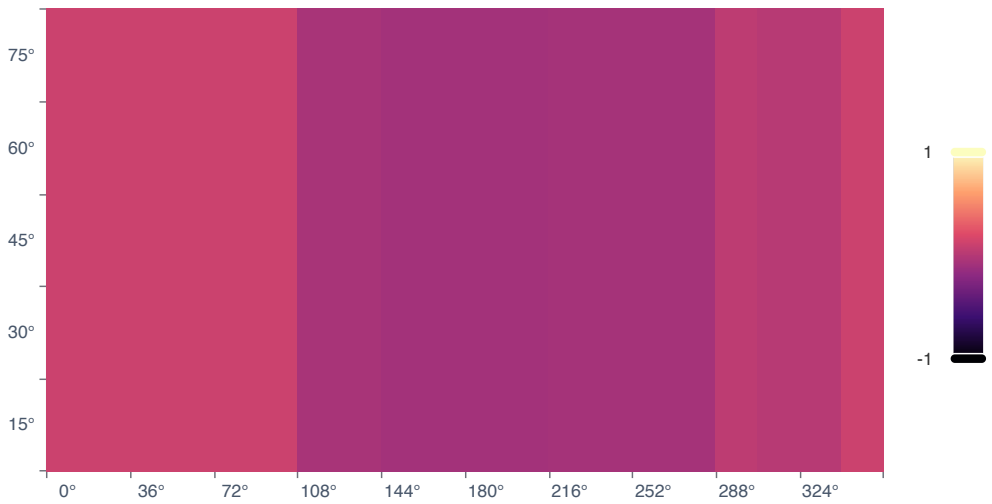

Figure 56: Gain vs (α_1, α_2).

DroppedCount	KeptSize	OriginalSize	ScanSteps	Step0Match	Step36
7	16	23	73	Democracy requires individual sacrifice.	Haste mak

Entropy	Margin	Phase	Step
3.251741509671787	0.7645083376282592	0°	0
1.9306301122846194	0.008642802078103959	180°	36
3.251741509671787	0.7645083376282591	360°	72

DistanceBand	MeanCorr
d[1,16]	-0.003448
d[17,64]	+0.002763
d[65,256]	+0.000970
d[257,511]	+0.003683

CorruptSim	Query	ScanSteps	
0.8223	Clean	73	Dem
0.8223	Corrupted (30% dropout): D__oc_ac_ re_u__es _ndi_idual sacrif_ce.	73	Dem

Alpha1	Base1	Base2	MidptGain	SnapAlpha	SnapGain
15°	0.1094	0.1330	0.1330	350°	0.1330
30°	0.1094	0.1330	0.1330	350°	0.1330
45°	0.1094	0.1330	0.1330	350°	0.1330
60°	0.1094	0.1330	0.1330	350°	0.1330
75°	0.1094	0.1330	0.1330	350°	0.1330

BestGain	Delta	Split	SteerMin	SuperAdditive
0.1330	+0.0000	192/320	0.1481	false
0.1330	+0.0000	224/288	0.1677	false
0.1417	+0.0087	256/256	0.2909	true
0.1330	+0.0000	288/224	0.2650	false
0.1417	+0.0087	320/192	0.1889	true

BestGain	Ceiling	Delta	Seed	Split	SuperAdditive
0.1417	0.1330	+0.0087	Democracy requires individual ...	T^2-256/256	true
0.1330	0.1330	+0.0000	Democracy requires individual ...	T^2-128/384	false
0.1330	0.1330	+0.0000	Democracy requires individual ...	T^2-384/128	false
0.1330	0.1330	+0.0000	Democracy requires individual ...	T^2-random	false
0.1417	0.1330	+0.0087	Democracy requires individual ...	T^2-energy	true
0.1479	0.1479	+0.0000	Knowledge is power.	T^2-256/256	false
0.1306	0.1479	-0.0172	Knowledge is power.	T^2-128/384	false
0.1306	0.1479	-0.0172	Knowledge is power.	T^2-384/128	false
0.1306	0.1479	-0.0172	Knowledge is power.	T^2-random	false
0.1306	0.1479	-0.0172	Knowledge is power.	T^2-energy	false
0.1914	0.1914	+0.0000	Nature does not hurry, yet eve...	T^2-256/256	false
0.1914	0.1914	+0.0000	Nature does not hurry, yet eve...	T^2-128/384	false
0.1914	0.1914	+0.0000	Nature does not hurry, yet eve...	T^2-384/128	false
0.1914	0.1914	+0.0000	Nature does not hurry, yet eve...	T^2-random	false
0.1914	0.1914	+0.0000	Nature does not hurry, yet eve...	T^2-energy	false

Alpha1	BestA1	BestA2	BestTorusGain	Ceiling	Delta	SuperAdditive
15°	50°	240°	0.1417	0.1330	+0.0087	true
30°	50°	240°	0.1417	0.1330	+0.0087	true
45°	50°	240°	0.1417	0.1330	+0.0087	true
60°	50°	240°	0.1417	0.1330	+0.0087	true
75°	50°	240°	0.1417	0.1330	+0.0087	true

Base1Max	Base2Max	BestGain	BestMatchB	
0.10938039770693596	0.13297593189643192	0.13297593189643192	A rolling stone gathers no moss.	De

6 Vision

Closed	ExactMatch	Generated	Image	Steps	TargetLen
False	False	5	Checkerboard	5	256
False	False	5	Horizontal Stripes	5	256
False	False	5	Vertical Stripes	5	256
False	False	5	Gradient	5	256
False	False	5	Cross	5	256

Closed	ExactMatch	Generated	Image	Steps	TargetLen
True	False	256	Checkerboard	11	256
True	False	256	Horizontal Stripes	11	256
True	False	157	Vertical Stripes	3	256
True	False	256	Gradient	7	256
True	False	256	Cross	9	256

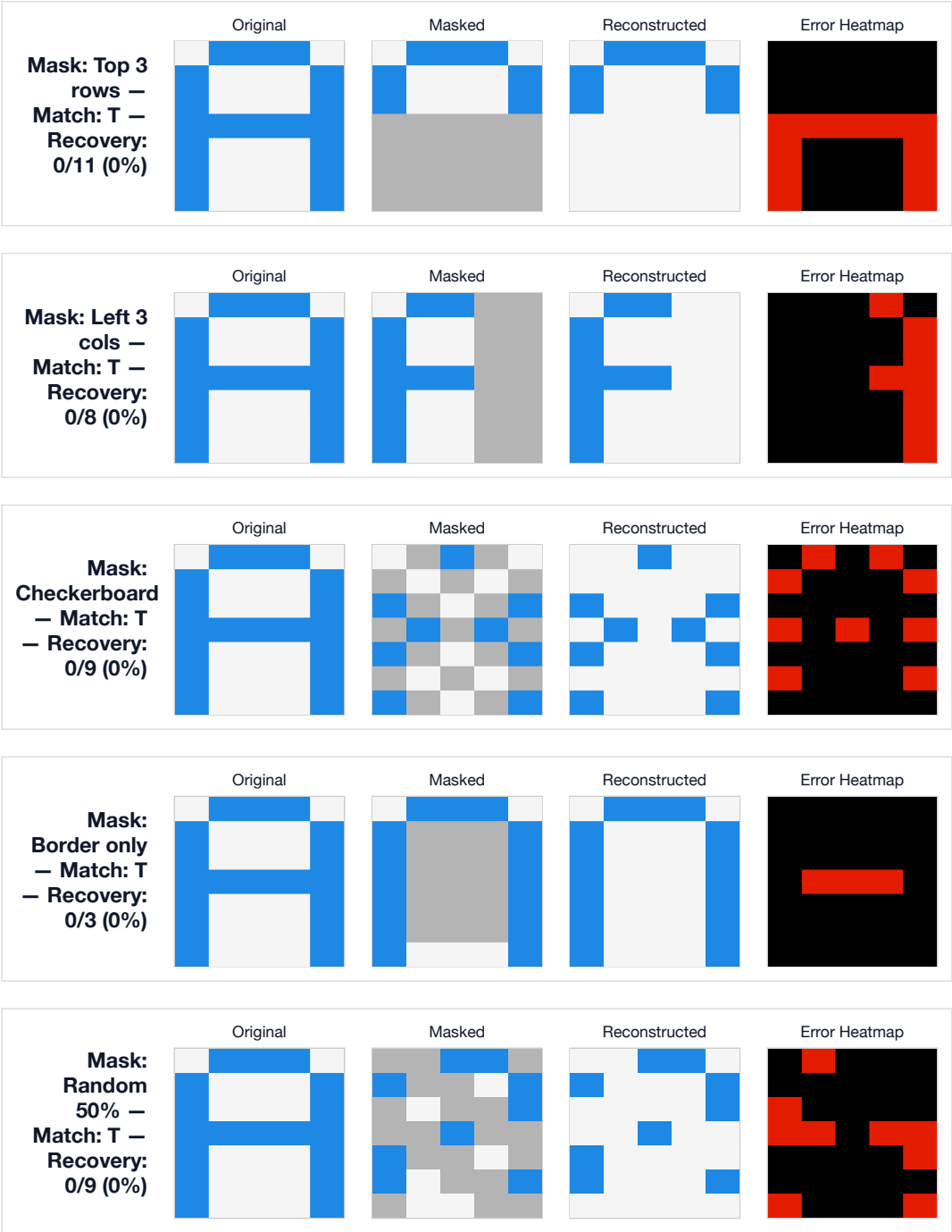

Figure 57: Letter 'A' reconstructed under 5 mask strategies using pure BestFill popcount resonance. Blue: known pixels. Green: recovered via ChordHole. Grey: masked region.

Figure 58: Image completions generated by topology routing directly over random holdout holes on CIFAR10.

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